

Analysis note for the paper
System size and energy dependence of near-side jet-like
correlations

By
Christine Nattrass (Yale), Jana Bielcikova (NPI), Jörn Putschke, Helen Caines

For the STAR collaboration

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Chapter 1

Particle selection and identification

1.1 Event selection

Data in this paper are from, $d + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV from RHIC's 2002 run, $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV and $\sqrt{s_{NN}} = 200$ GeV from RHIC's fourth run (2004) and $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 62$ GeV and $\sqrt{s_{NN}} = 200$ GeV from RHIC's fifth run (2005.) For all systems and energies, the vertex is limited to within 30 cm of the center of the TPC along the beam axis. The further an event is from the center of the TPC the lower the geometric acceptance; a 30 cm cut allows reasonably uniform geometric acceptance. The production used and the number of good events is listed in Table 1.1 for all data sets.

The centrality bins used in this analysis and their corresponding reference multiplicities, N_{bin} , and N_{part} are given in Appendix C.

Events from $Au + Au$ collisions have a more clearly defined vertex because there are more tracks produced in an $Au + Au$ collision which can be used to reconstruct the vertex position. The 80% most central events were usable in $Au + Au$ collisions. Fewer tracks are produced in $Cu + Cu$ collisions and the luminosity was significantly higher in 2005 than in 2006. The higher luminosity meant that particles from a previous event could still be traversing the TPC when a new event triggered the detector to begin reading out data, a problem called pile-up. For higher multiplicities, it is still possible to determine the primary vertex, but the lower the multiplicity the harder it

Table 1.1: Information about the productions used in the analysis

collision system and energy	production used		number of events
$d + Au \sqrt{s_{NN}} = 200 \text{ GeV}$	P03ih		1M
$Cu + Cu \sqrt{s_{NN}} = 62 \text{ GeV}$	P05id		24M
$Cu + Cu \sqrt{s_{NN}} = 200 \text{ GeV}$	P06ib		38
$Au + Au \sqrt{s_{NN}} = 62 \text{ GeV}$	P04id and—P04ie		8M
$Au + Au \sqrt{s_{NN}} = 200 \text{ GeV}$	P05ic	24M central, 25M minimum bias	

is to determine where the vertex from the event which triggered the detector is. For this reason only the 60% most central data from these events was usable. For more peripheral events, the vertex which triggered the event could not be found in more than 2% of events, in large part because vertices from pile-up events usually had a greater multiplicity. Full details of the separation of the vertices from the triggered event and from pile-up events can be found in [1].

1.2 Selection of events and charged tracks

All particles are restricted to $|\eta| < 1.0$. All tracks used in these studies are required to have at least 15 hits in the TPC associated with the track. Tracks are restricted to a DCA $< 1.0 \text{ cm}$, limiting contributions from decay particles.

Unidentified hadron efficiency

To determine the efficiency for reconstructing tracks, the response to the particle as it traverses the STAR detector is simulated. For unidentified hadrons, the response of

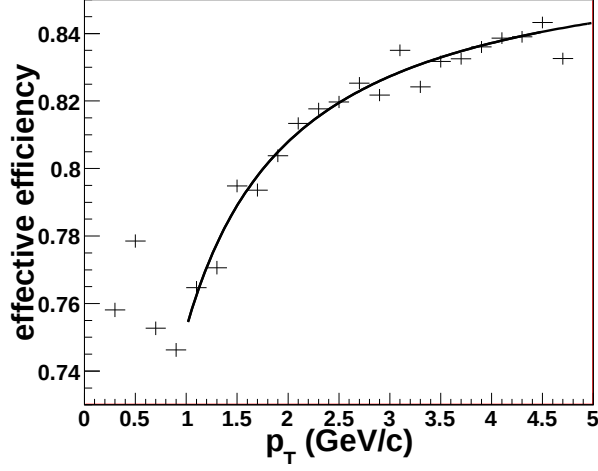


Figure 1.1: The effective efficiency, including acceptance effects, for reconstructing unidentified hadrons in 0-10% central $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV. The curve is a fit to the data.

the detector is simulated for pions. There are significant contributions from protons in the data, however, the efficiency for the reconstruction of pions and protons is similar for particles with $p_T > 1$ GeV/c. The reconstructed track must have at least 10 of the simulated hits for the simulated track to be considered reconstructed. The ratio of the number of reconstructed tracks to the number of simulated tracks gives the effective efficiency, including acceptance effects. The effective reconstruction efficiency was determined for tracks meeting the quality requirements applied in the analysis and was determined as a function of p_T and centrality for each data set. This was fit to a curve $a \exp((\frac{b}{p_T})^c)$. An example of the effective efficiency for the reconstruction of unidentified hadrons in 0-10% central $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV is shown in Figure 1.1. For $d + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV, a constant efficiency of 0.89 was used. The parameters for the fits to the efficiency as a function of p_T for each centrality bin for every other data set are in Appendix ??.

Chapter 2

Correlation Method

2.1 Correlation technique

In a given event, a high- p_T track, called the trigger particle, is selected and the distribution of all lower momentum particles, called associated particles, relative to that track in $\Delta\eta$ and $\Delta\phi$ is determined. Tracks are selected as described in Chapter 1, with no additional restrictions on the high momentum track other than the p_T . Multiple tracks in a single event may be counted as trigger particles. The data are symmetric about $\Delta\phi = 0$ and $\Delta\eta = 0$ because the sign of $\Delta\phi$ and $\Delta\eta$ are determined by an arbitrary choice of coordinates. Since many of these measurements are limited by statistics, the data are reflected about $\Delta\phi = 0$ and $\Delta\eta = 0$ to minimize statistical fluctuations. The single track efficiency, described in Chapter 1, is dependent on particle type, collision system and energy, p_T , and centrality and the correction for associated particles is applied on a track-by-track basis. The final distribution is normalized by the number of trigger particles, so no correction for the efficiency of reconstructing the trigger particle is necessary. Two corrections for the acceptance of track pairs in $\Delta\phi$ and $\Delta\eta$, described below, are applied. This is done for the full sample of events to give the average distribution of associated particles relative to a high- p_T trigger particle $\frac{1}{N_{trigger}} \frac{d^2N}{d\Delta\phi d\Delta\eta}$.

Two key features of the STAR TPC affect the geometric acceptance of track pairs. Tracks in the TPC can only be resolved well for $-1 < \eta < 1$ because the lower

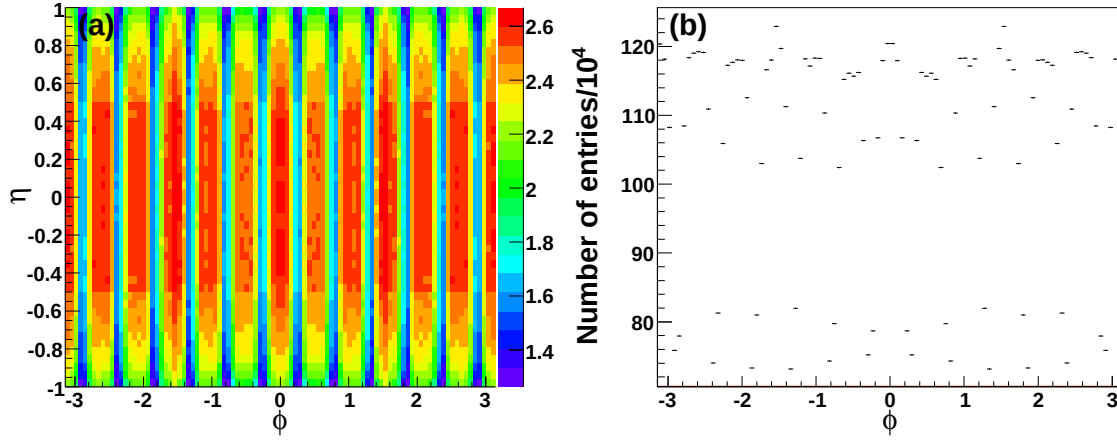


Figure 2.1: Distribution of tracks in (a) ϕ and η and (b) ϕ in $Cu + Cu$ events at $\sqrt{s_{NN}} = 200$ GeV for $1.5 \text{ GeV}/c < p_T < 6.0 \text{ GeV}/c$. Note the suppressed zero.

geometric acceptance leads to tracks with fewer hits in the TPC, so only tracks within $-1 < \eta < 1$ are used for this analysis. Track pairs with $\Delta\eta \approx 0$ are nearly always accepted, however, track pairs with $|\Delta\eta| \approx 2$ are only accepted if the tracks are near opposite edges of the detector. Therefore acceptance is roughly 100% near $\Delta\eta \approx 0$ but roughly 0% near $|\Delta\eta| \approx 2$. This leads to an acceptance in $\Delta\eta$ with a triangular shape.

In addition, as discussed in Chapter ??, there are gaps between each of the 12 TPC sectors. This leads to holes in the acceptance in azimuth. Most tracks will have hits in multiple sectors, however, the loss of hits leads to a decrease in the reconstruction efficiency for tracks at the angles of the sector boundaries. A sample distribution of tracks reconstructed in the TPC is shown in Figure 2.1 showing the lower efficiency at the TPC sector boundaries.

To correct for these effects, the distributions of tracks for both the associated and the trigger particles are recorded. A random trigger (ϕ, η) is chosen from this distribution of trigger particles and a random associated (ϕ, η) is chosen from the distribution of associated particles. In this way a simulated distribution of uncorrelated track pairs that includes detector acceptance effects is attained. This distribution is normalized to 1 at the peak at $\Delta\eta \approx 0$ to give an efficiency for finding a pair of tracks as a function of $(\Delta\phi, \Delta\eta)$. A sample of the efficiency due to detector effects is shown

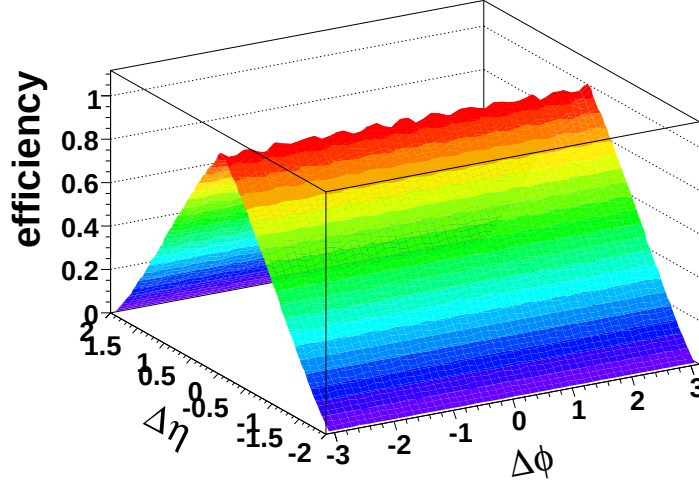


Figure 2.2: Sample efficiency due to detector effects for $Cu + Cu$ events at $\sqrt{s_{NN}} = 200$ GeV with $1.5 \text{ GeV}/c < p_T^{associated} < p_T^{trigger}$ and $3.0 < p_T^{trigger} < 6.0 \text{ GeV}/c$

in Figure 2.2 and the effects of the geometric acceptance are evident.

2.1.1 Track merging correction

If two tracks have small separations in pseudorapidity and azimuth, they are more difficult to resolve because hits from each particle may not be resolved by the TPC. Therefore, if the particles have nearly identical momenta, their tracks are not distinguished. This effect is called track merging and causes an artificial dip in the raw correlations centered at $(\Delta\phi, \Delta\eta) = (0,0)$. Another similar effect is evident in the TPC from the crossing of high- p_T tracks. While the hits are not lost in this case, one track will lose hits near the crossing point and this particle's track gets split into two shorter tracks. Shorter tracks are less likely to meet the track quality cuts in Chapter 1 because they have fewer hits. This effect causes artificial dips near $(\Delta\phi, \Delta\eta) = (0,0)$ but slightly displaced in $\Delta\phi$. The location of these dips in $\Delta\phi$ is dependent on the relative helicities of the trigger and associated particles. The helicity is given by

$$h = -\frac{qB}{|qB|} \quad (2.1)$$

where q is the charge of the particle and B is the magnetic field [2].

A method for correcting high- p_T triggered di-hadron correlations was developed in [2, 3], building on a method for correcting for track merging in studies of Hanbury-Brown-Twiss correlations [4].

Correction for unidentified hadrons

Figure 2.3 shows a sample of raw di-hadron correlations between unidentified hadrons for all four helicity combinations in $Au + Au$ events at $\sqrt{s_{NN}} = 62$ GeV. When the helicities of the trigger and associated particles are the same, the percentage of overlapping hits is greater. Track pairs are lost whether the pair is part of the combinatorial background or part of the signal so the magnitude of the dip is greater when the background is greater. $Au + Au$ collisions were chosen to demonstrate the effect because the background is largest for these collisions, making the effect easy to see.

The number of hits shared by two tracks is calculated based on the trajectory of the track and if more than 10% of hits are shared between two tracks, the tracks are considered merged and the track pair is discarded. To get an acceptance correction for this effect, the trigger particle is mixed with associated particles from another event. This procedure is called event mixing. The trigger and associated particles from separate events will have no real correlations so any structures in the resulting $(\Delta\phi, \Delta\eta)$ distribution will result only from detector acceptance effects, including track merging. In the mixed events, the origin of the associated track must be shifted to the new vertex so that the percentage of merged hits can be calculated accurately. In addition, mixed events are required to have vertices within 2 cm of each other along the beam axis in order to ensure similar geometric acceptance in both events. The reference multiplicities of both events are required to be within 10. The calculation of the number of hits which may be shared by the tracks does not describe the performance of the TPC perfectly and some track pairs which were calculated to share more than 10% of their hits are reconstructed. These pairs are discarded even if they were reconstructed so that the percentage of merged track pairs is the same in the data and the mixed events. The mixed events are then used to determine an

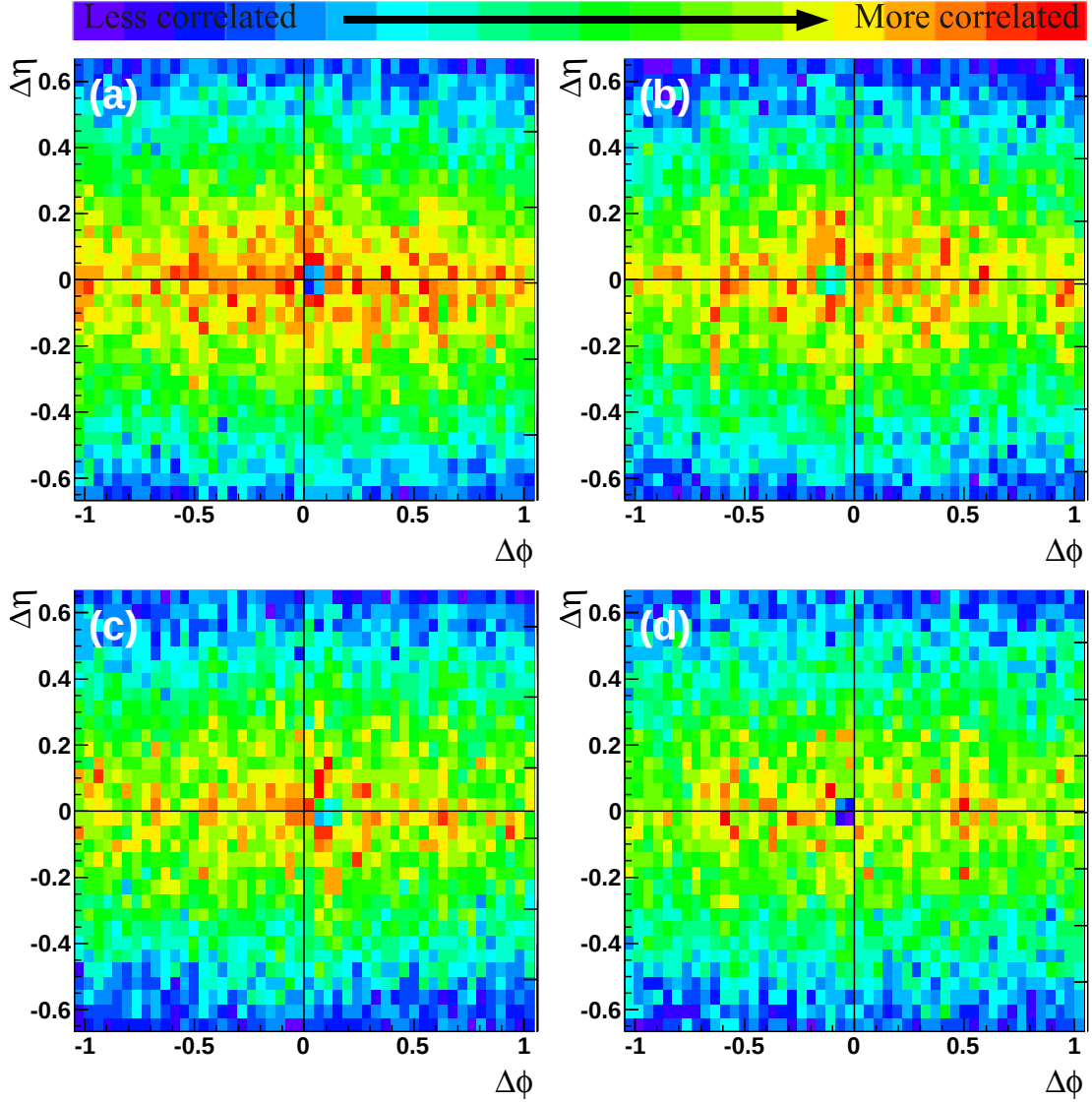


Figure 2.3: Raw correlations for (a) trigger particle helicity 1 and associated particle helicity 1 (b) trigger particle helicity 1 and associated particle helicity -1 (c) trigger particle helicity -1 and associated particle helicity 1 (d) trigger particle helicity -1 and associated particle helicity -1 in 0-40% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV for $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c and $3.0 < p_T^{trigger} < 6.0$ GeV/c.

acceptance correction as a function of $\Delta\phi$ and $\Delta\eta$ for track pairs. Most of the dip is corrected away when the data are divided by the efficiency from mixed events [2, 3].

While the method discussed in Section 2.1 corrects for the geometric acceptance of the detector, this method corrects for both the geometric acceptance and for track merging. This method is much more CPU-intensive and therefore was only applied for small $(\Delta\phi, \Delta\eta)$.

Figure 2.4(a) shows the data and Figure 2.4(b) shows the mixed events for a sample correlation. The dip has a different shape when the helicities of the tracks are the same and when they are opposite. To do the correction the data and mixed events are rotated so that the dip is in the same location. The same helicity data are divided by the acceptance correction from the same helicity events and the opposite helicity data are divided by the acceptance correction from the opposite helicity mixed events. These data are then added after the acceptance correction. A small residual dip remains because mixed events do not recreate the dip perfectly. The remaining dip is corrected for using the symmetry of the correlations. Since the data are symmetric about $\Delta\phi = 0$, the data on the same side as the dip are discarded and replaced by the data on the side without the dip. Then the data are reflected about $\Delta\eta = 0$ and added to the unreflected data to minimize statistical fluctuations.

This method is only applied to $|\Delta\phi| < 1.05$ and $|\Delta\eta| < 0.67$ because it is CPU-intensive and track merging only affects small $\Delta\phi$ and small $\Delta\eta$. For large $\Delta\phi$ and $\Delta\eta$, the method described in Section 2.1 is applied. Figure 2.4(c) shows the data after the track merging correction and Figure 2.4(d) shows the data using the method described in Section 2.1 - without the track merging correction - for all helicity combinations for 0-40% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV.

2.2 Separation of the jet-like correlation and the *Ridge*

Figure 2.5 shows a sample correlation after detector acceptance and track merging corrections. To study the near-side of di-hadron correlations, the two components are separated and the yield, number of particles in each within limits on $p_T^{associated}$ and $p_T^{trigger}$, is studied. As discussed in Chapter ??, the *Ridge* was previously observed

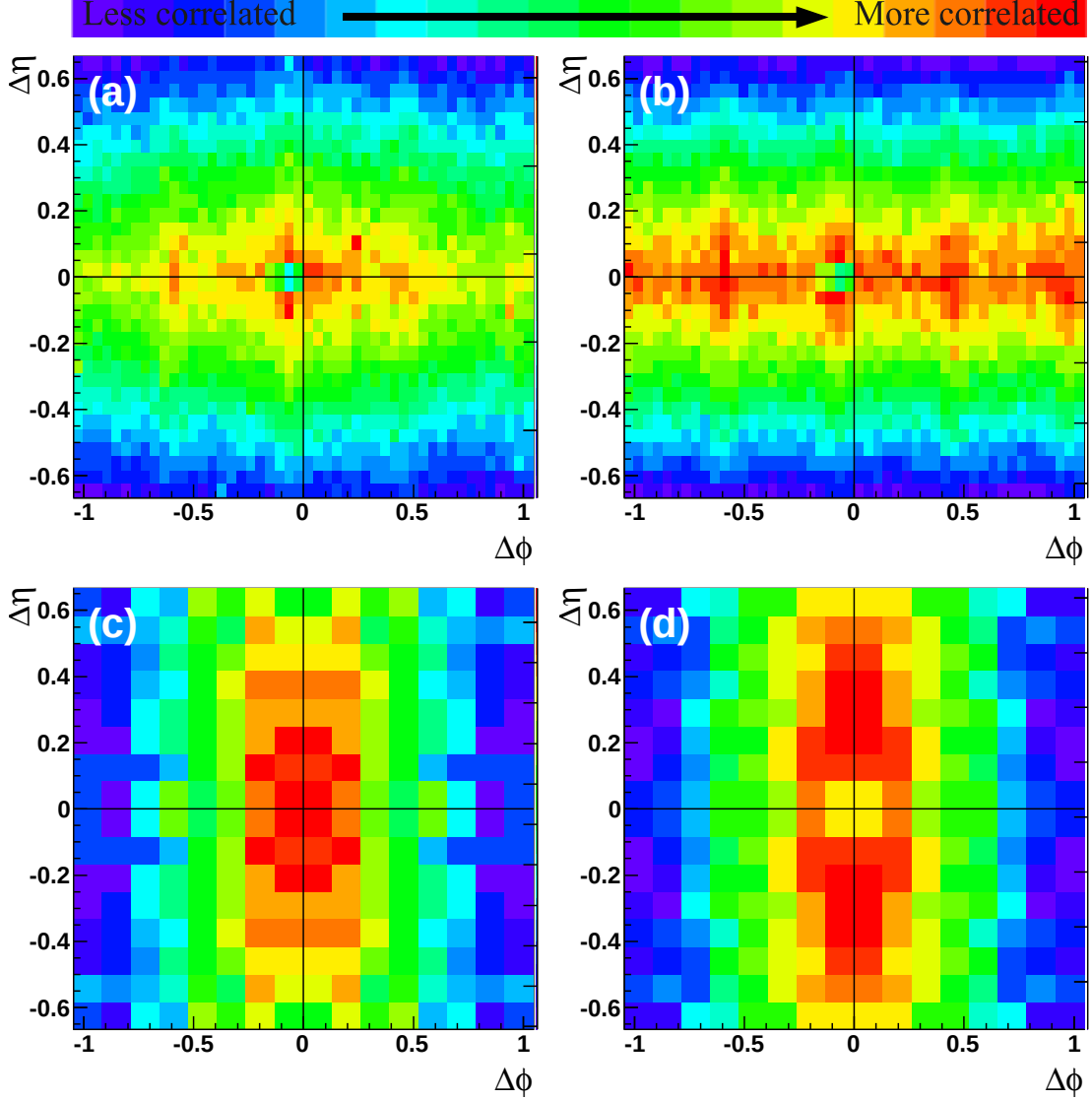


Figure 2.4: The region around the dip for (a) data (b) mixed events (c) data after track merging correction (d) data using the method described in Section 2.1 (no track merging correction) in 0-40% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV for $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c and $3.0 < p_T^{trigger} < 6.0$ GeV/c.

to be roughly independent of $\Delta\eta$ within the acceptance of the STAR TPC while the jet-like correlation is confined to $|\Delta\eta| < 0.75$.

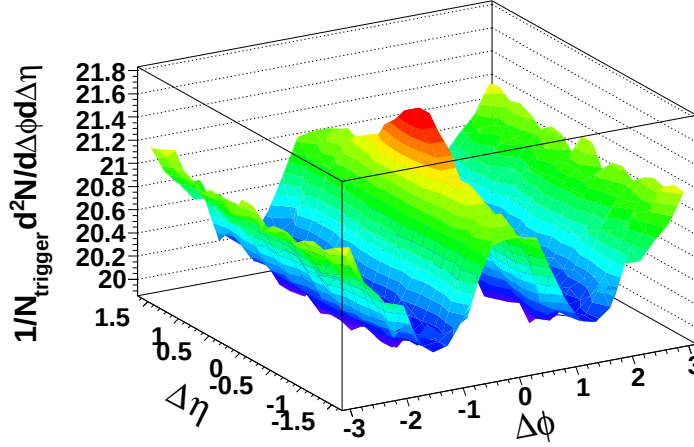


Figure 2.5: Sample correlation after acceptance corrections for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 \text{ GeV/c} < p_T^{associated} < p_T^{trigger}$ from 0-40% most central Au + Au collisions at $\sqrt{s_{NN}} = 62$ GeV.

2.2.1 Extraction of the jet-like yield

The raw correlation consists of a combinatorial background modulated by elliptic flow, v_2 , and the signal, which consists of the jet-like correlation and the *Ridge* on the near-side. There are two methods used to extract the jet-like yield from the raw correlation. The first method uses different projections in $\Delta\phi$, the fact that the jet-like correlation is not present at large $\Delta\eta$, and the fact that the combinatorial background and the *Ridge* are both roughly independent of $\Delta\eta$ to isolate the jet-like correlation as a function of $\Delta\phi$. The second method involves taking a projection in $\Delta\eta$ for $-1 < \Delta\phi < 1$ and subtracting a constant background to determine the jet-like correlation as a function of $\Delta\eta$. Elliptic flow, v_2 , does not lead to a systematic error on the jet-like yield using either of these methods because v_2 is roughly independent of $\Delta\eta$ within the STAR acceptance [5, 6]. Previous studies in Au + Au collisions at $\sqrt{s_{NN}} = 200$ GeV have found these methods to be consistent at high- p_T [7], with some indications that there may be greater deviations at lower p_T [8]. Both methods are described in greater detail below.

$\Delta\phi$ method

To determine the jet-like yield, Y_{Jet} , the projection of the distribution of particles $\frac{d^2N}{d\Delta\phi d\Delta\eta}$ is taken in two different ranges in pseudorapidity:

$$\frac{dY_{ridge}}{d\Delta\phi} = \frac{1}{N_{trigger}} \int_{-1.75}^{-0.75} \frac{d^2N}{d\Delta\phi d\Delta\eta} d\Delta\eta + \frac{1}{N_{trigger}} \int_{0.75}^{1.75} \frac{d^2N}{d\Delta\phi d\Delta\eta} d\Delta\eta \quad (2.2)$$

$$\frac{dY_{jet+ridge}}{d\Delta\phi} = \frac{1}{N_{trigger}} \int_{-0.75}^{0.75} \frac{d^2N}{d\Delta\phi d\Delta\eta} d\Delta\eta \quad (2.3)$$

where the former contains only the *Ridge* and the latter contains both the jet-like correlation and the *Ridge*. The jet-like yield, Y_{Jet} , is determined using bin counting in the region $-1 < \Delta\phi < 1$:

$$Y_{jet}^{\Delta\phi} = \int_{-1}^1 \left(\frac{dY_{jet+ridge}}{d\Delta\phi} - \frac{0.75}{1} \frac{dY_{ridge}}{d\Delta\phi} \right) d\Delta\phi. \quad (2.4)$$

The factor in front of the second term is the ratio of the $\Delta\eta$ width in the region containing the jet-like correlation and the *Ridge* to the width of the region containing only the *Ridge*. In practice because the data have been reflected about $\Delta\phi = 0$ and $\Delta\eta = 0$, the projections in $\Delta\phi$ and $\Delta\eta$ and the bin counting are done over half of the range and then scaled up to ensure that the errors are correct.

$\Delta\eta$ method

In this method the projection of the distribution of particles $\frac{d^2N}{d\Delta\phi d\Delta\eta}$ on the near-side is taken:

$$\frac{1}{N_{trigger}} \frac{dN}{d\Delta\eta} = \frac{1}{N_{trigger}} \int_{-1}^1 \frac{d^2N}{d\Delta\phi d\Delta\eta} d\Delta\phi. \quad (2.5)$$

This contains both a background due to combinatorics and the *Ridge*, both of which are independent of $\Delta\eta$. The background is determined first by fitting a Gaussian signal plus a constant background, A, to Equation 2.6 and then using the background from that for background subtraction. If statistics permitted, the background could

be determined from $|\Delta\eta| > 0.75$, however, many of the correlations presented here have limited statistics and using a fit of the whole region to determine the background prevents statistical fluctuations from skewing the background subtraction. The jet-like yield is:

$$Y_{jet}^{\Delta\eta} = \frac{1}{N_{trigger}} \int_{-1}^1 \left(\frac{d^2N}{d\Delta\phi d\Delta\eta} - A \right) d\Delta\phi. \quad (2.6)$$

The yield is determined by bin counting in the range $|\Delta\eta| < 1$. To get the proper errors, the projection and the bin counting are also done over half of the range and scaled because the data have been reflected. The statistical error on A from the fit is used to determine the error on the final yield by adding it in quadrature to the statistical error from bin counting. The magnitudes of these two errors are comparable.

Figure 2.6 shows the jet-like correlation from both the $\Delta\phi$ and the $\Delta\eta$ method after background subtraction for all systems and energies using $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 \text{ GeV/c} < p_T^{associated} < p_T^{trigger}$, the limits used for most studies presented here. For all collision systems and energies the jet-like correlation is within $|\Delta\eta| < 0.75$.

2.2.2 Extraction of the *Ridge* yield

There is a large combinatorial background which is dependent on $\Delta\phi$. The raw signal has a background due to particles correlated indirectly with each other in azimuth due to their correlation with the reaction plane. This random background is estimated by

$$\frac{dY_{bkgd}}{d\phi} = B(1 + 2\langle v_2^{trigger} \rangle \langle v_2^{associated} \rangle \cos(2\Delta\phi)) \quad (2.7)$$

where v_2 is the second order harmonic in a Fourier expansion of the momentum anisotropy relative to the reaction plane [9]. The derivation of this parameterization of the background is dependent on the assumption that di-jet production is not correlated with the reaction plane so that the correlation between particles in the jet and random particles in event are not correlated in azimuth through a mutual correlation with the reaction plane. Systematic errors come from the errors on B, $\langle v_2^{trigger} \rangle$ and $\langle v_2^{associated} \rangle$. v_2 is determined from independent measurements. It is assumed that v_2

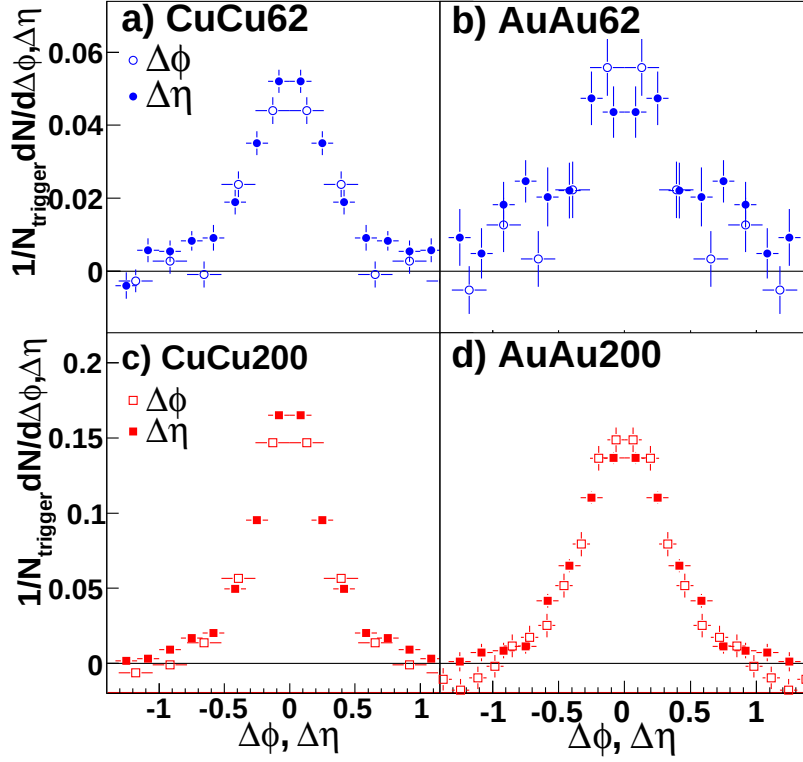


Figure 2.6: Sample jet-like correlations for $3.0 < p_T^{trigger} < 6.0$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ from both the $\Delta\phi$ and the $\Delta\eta$ method for (a) 0-60% central $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 62$ GeV (b) 0-80% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV (c) 0-60% central $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV and (d) 0-80% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

is the same for events with a trigger particle as for minimum bias events and that v_2 is independent of $\Delta\eta$. For each data set $v_2(p_T)$ was fit in centrality bins to determine $\langle v_2^{trigger} \rangle$ and $\langle v_2^{associated} \rangle$.

Measurements of v_2

The best method for the determination of v_2 depends on the collision system. Since v_2 is a momentum anisotropy in azimuth, the tracks produced in an event can be used to reconstruct the reaction plane. v_2 in Equation 2.7 is the azimuthal momentum anisotropy due to the collective flow of particles, however, there are other sources of azimuthal momentum anisotropy. For particles above 1 GeV/c, jets are the largest

known non-flow source of momentum anisotropy. Various methods for measuring v_2 have different non-flow contributions. The systematic error is determined by comparing these different methods. Generally, the higher the multiplicity of an event and the greater the eccentricity of the overlap region, the easier it is to reconstruct the event plane. Therefore that the event plane is more difficult to reconstruct in $Cu + Cu$ collisions.

For $Cu + Cu$ collisions, the nominal for v_2 was taken from a measurement of v_2 using tracks from the FTPCs for reconstruction of the event plane. v_2 measured in $Cu + Cu$ using the event plane from the FTPCs is shown as a function of p_T and centrality in Figure 2.7 [10]. These data were fit to a parameterization which was used for the subtraction of v_2 using Equation 2.7. The fit parameters are given in Table 2.1. The statistical errors on the fits are taken into account and combined with the systematic errors (5% of v_2 for 0-40% central collisions and 10% of v_2 for 40-60% central collisions.) For $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV the v_2 for 40-50% central collisions is used for 50-60% central collisions because the errors in the most peripheral bin were large enough that these data could not be fit and data from the two centralities were consistent. The event plane reconstructed from tracks measured in both of the FTPCs is expected to be less biased by jets since tracks in the FTPCs are separated by more than two units of pseudorapidity. Particles produced in di-jets in $p + p$ collisions do not exhibit long range correlations in pseudorapidity.

v_2 and the systematic errors on v_2 in $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV were taken from [11]. The dominant systematic errors were determined by comparing the reaction plane method to the four-particle cumulant method. The four-particle cumulant method determines v_2 by looking at correlations between four particles, which significantly reduces its sensitivity to non-flow. v_2 and the magnitude of the negative systematic error on v_2 in $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV as a function of p_T and centrality from [11] are shown in Figure 2.8. These were fit to a parameterization of v_2 and the fit parameters are given in Table 2.2. Statistical errors on the fits are also neglected for $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV since the systematic errors on v_2 are several times larger than the statistical errors. The positive systematic error in $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV is a constant, 0.003, for all centralities and

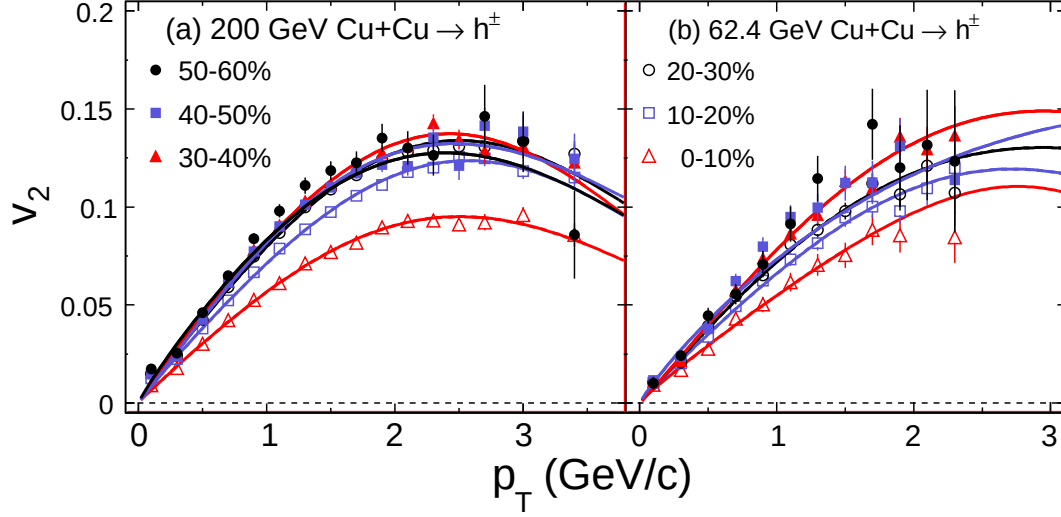


Figure 2.7: v_2 measured using the event plane from the FTPCs in $Cu + Cu$ collisions at (a) $\sqrt{s_{NN}} = 200$ GeV and (b) $\sqrt{s_{NN}} = 62$ GeV. Fit parameters are given in Table 2.1

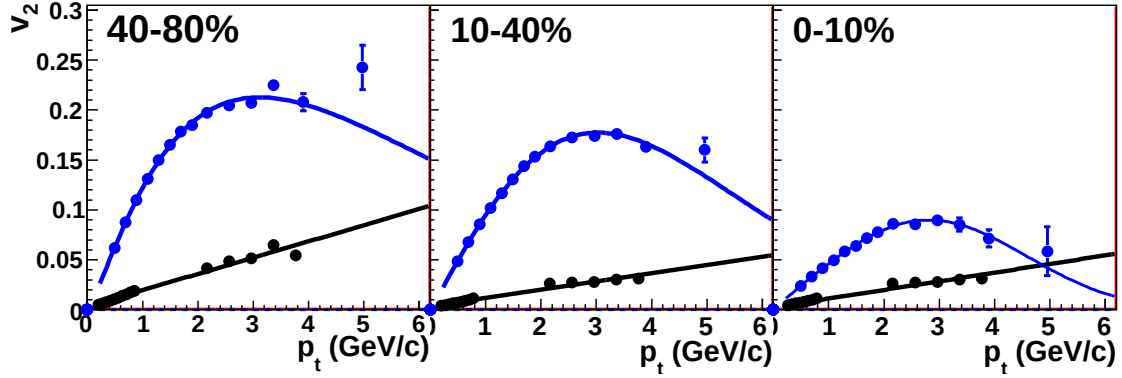


Figure 2.8: v_2 (blue) and the negative systematic error on v_2 (black) in $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV. Fit parameters are given in Table 2.2.

p_T .

v_2 and the systematic errors on v_2 in $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV were taken from http://www.star.bnl.gov/protected/jetcorr/mvl/v2_compare/auau_compare.html,

Table 2.1: Fit parameters from fits of v_2 measured using the event plane reconstructed using the FTPC in $Cu+Cu$ shown in Figure ?? and Figure ?? to $v_2(p_T) = ap_T^b e^{-(p_T/c)^d} \exp(\frac{b}{p_T})$

centrality	200 GeV			
	a	b	c	d
0-10%	0.0594 ± 0.0011	0.9541 ± 0.0192	3.6778 ± 0.0936	2.3449 ± 0.2438
10-20%	0.0735 ± 0.0008	0.9368 ± 0.0127	3.8002 ± 0.0779	2.5779 ± 0.2030
20-30%	0.0836 ± 0.0011	0.9599 ± 0.0141	3.6625 ± 0.0710	2.3440 ± 0.1711
30-40%	0.0863 ± 0.0013	0.9422 ± 0.0163	3.5843 ± 0.0844	2.5421 ± 0.2262
40-50%	0.0910 ± 0.0038	1.0349 ± 0.0364	3.3741 ± 0.1583	1.8515 ± 0.2546
50-60%	0.0888 ± 0.0022	0.8928 ± 0.0247	3.6000 ± 0.0000	2.1461 ± 0.2273
centrality	62 GeV			
	a	b	c	d
0-10%	0.0550 ± 0.0029	0.9076 ± 0.0653	4.0000 ± 0.0000	4.0291 ± 2.7322
10-20%	0.0684 ± 0.0033	0.9539 ± 0.0514	4.0000 ± 0.0000	2.3958 ± 0.6978
20-30%	0.0787 ± 0.0051	1.0126 ± 0.0500	4.0000 ± 0.0000	1.7281 ± 0.4684
30-40%	0.0843 ± 0.0067	1.0519 ± 0.0666	4.0000 ± 0.0000	1.8512 ± 0.6561
40-60%	0.0825 ± 0.0055	0.9039 ± 0.0484	5.6000 ± 0.0000	1.2375 ± 0.2896

Table 2.2: Fit parameters from fits of v_2 from [11] to $v_2(p_T) = ap_T^b e^{-(p_T/c)^d} \exp(\frac{b}{p_T})$ and of the negative systematic error on v_2 to $mp_T + b$

centrality	v_2				systematic error	
	a	b	c	d	m	b
0-10%	0.205	1.30	2.08	0.90	0.0163	0.0035
10-40%	0.105	1.04	4.03	1.67	0.0082	0.0038
40-80%	0.047	0.96	4.11	2.72	0.0087	0.0026

where the new parametrization which was agreed upon in the jetcorr PWG in 2009 is described. The v_2 parametrization as a function of p_T and centrality is shown in Figure 2.9. The upper limit is given by the fit to the measurement of $v_2\{FTPC\}$ and the lower limit is the fit to the four-particle cumulant $v_2\{4\}$. For the most peripheral and central bin, the lower limit is determined from an extrapolation. The functional form of the parametrization is $v_2(p_T, cent) = A \cdot p_T^n \cdot e^{B \cdot p_T}$. The values of the parameter B is -0.445 (upper limit) and -0.5 (lower limit). The parameter n depends on

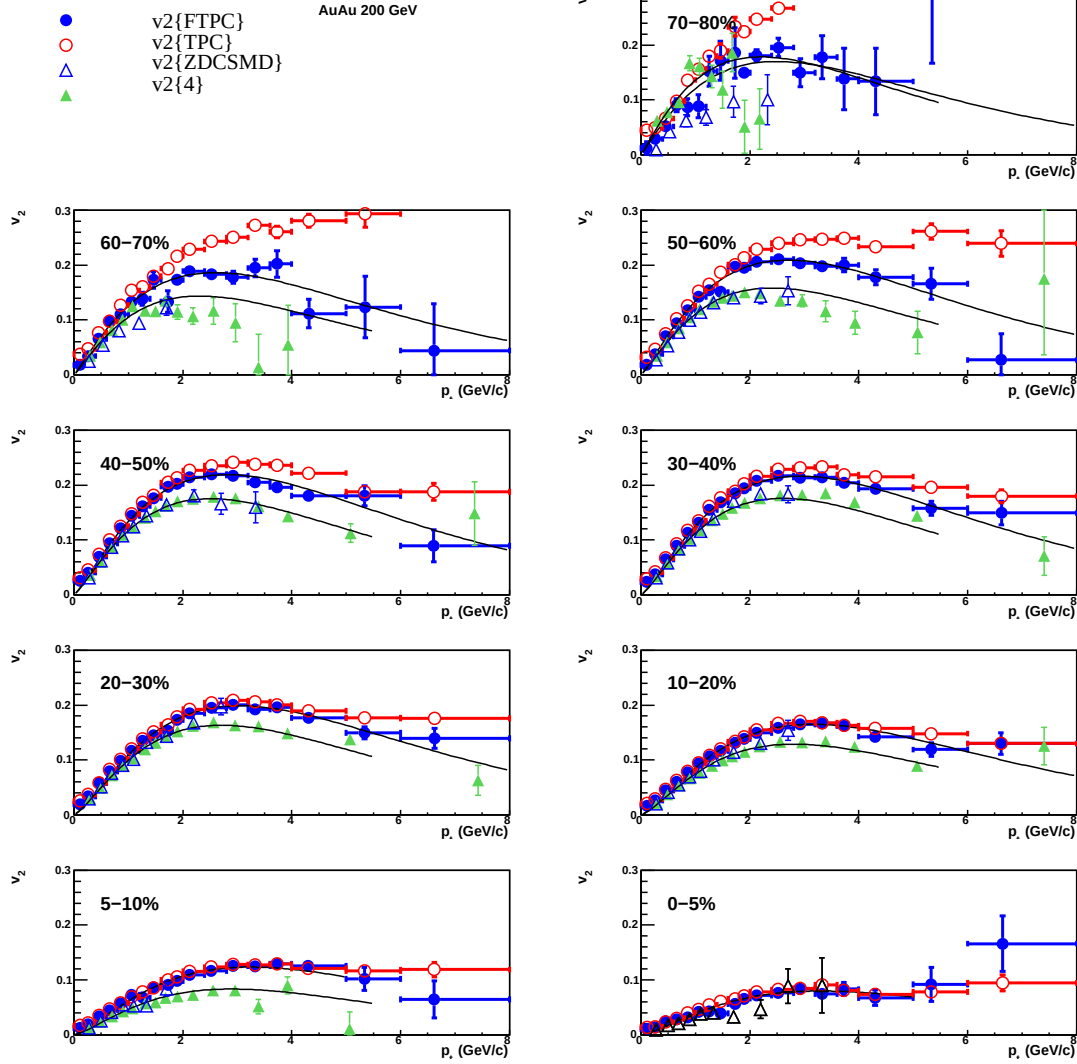


Figure 2.9: Symbols represent v_2 in $Au+Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV determined by various methods as described in the legend. The black lines are the upper and lower systematic boundaries on v_2 using the fit function $v_2(p_T, cent) = A \cdot p_T^n \cdot e^{B \cdot p_T}$. The values of the parameters are $B=-0.445$ (upper curve), $B=-0.5$ (lower curve), and $n=1.48-0.5 \cdot cent$.

centrality as $n=1.48-0.5 \cdot cent$.

B is determined using the Zero Yield At Minimum (ZYAM) method [12, 6]. This method fixes B by assuming that there is no signal at the minimum of the correlation

in azimuth. For measurements with low statistics the statistical error on the minimum data point is a significant contribution to the error on the *Ridge* yield. This error is added in quadrature to the statistical error.

To determine the yield of the *Ridge*, Y_{Ridge} , the di-hadron correlation $\frac{d^2N}{d\Delta\phi d\Delta\eta}$ is projected over the entire $\Delta\eta$ region. To minimize the effects of statistical fluctuations in the determination of the background, $\frac{dY_{bgd}}{d\phi}$, Y_{Jet} is subtracted after the projection in $\Delta\eta$:

$$Y_{Ridge} = 1/N_{trigger} \int_{-1.75}^{1.75} \int_{-1}^1 \left(\frac{d^2N}{d\Delta\phi d\Delta\eta} - \frac{dY_{bgd}}{d\phi} \right) d\Delta\phi d\Delta\eta - Y_{Jet}^{\Delta\eta}. \quad (2.8)$$

The integration over $\Delta\phi$ is done by bin counting. $Y_{Jet}^{\Delta\eta}$ is used for the subtraction of Y_{Jet} because it is less sensitive to the assumption that the jet-like correlation is contained within $|\Delta\eta| < 0.75$. Sample correlations in azimuth after the projection over $\Delta\eta$ are shown in Figure 2.10 for all data sets. These correlations show the background level and shape for the high and low bounds on v_2 .

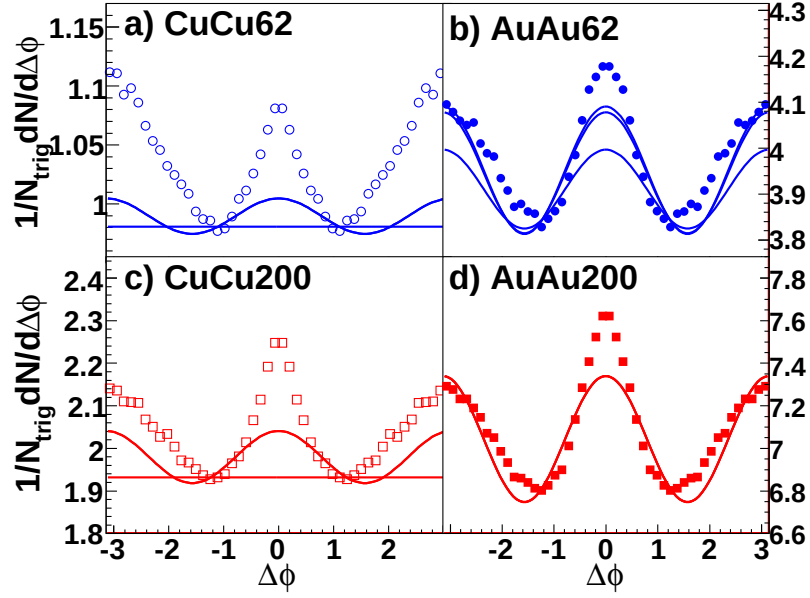


Figure 2.10: Sample correlations in $\Delta\phi$ integrated over $|\Delta\eta| < 1.75$ for $3.0 < p_T^{trigger} < 6.0$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ from (a) 0-60% central $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 62$ GeV (b) 0-80% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV (c) 0-60% central $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV and (d) 0-80% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV showing the background level and shape for high and low bounds on v_2 from each data set.

Chapter 3

Results

3.1 The jet-like correlation

If the jet-like correlation is produced by fragmentation, no dependence on collision system or system size would be expected. The particle ratios of the jet-like correlation should be consistent with what was observed in $p + p$ collisions since $p + p$ collisions are expected to be dominated by jet production at high- p_T . The jet-like yield should increase with $p_T^{trigger}$ because higher p_T trigger particles come on average from higher energy jets and the spectra of particles in the jet-like correlation should be harder than that observed in the inclusive. Studies of the jet-like yield as a function of collision system and energy were done for unidentified hadrons for $Cu + Cu$ and $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV and $\sqrt{s_{NN}} = 200$ GeV to determine whether the data are consistent with the jet-like correlation being produced by fragmentation.

3.1.1 Flavor and system dependence

Differences between baryons and mesons have been observed in R_{AA} and in v_2 and models such as Recombination hypothesizes novel methods for production of baryons and mesons in $A + A$ collisions to explain the observed differences. If particles in jets are produced by vacuum fragmentation, baryon/meson ratios would be expected to be comparable to $p + p$. Jets with leading baryons may have different properties than

jets with leading mesons. Particles produced by mechanisms such as Recombination would have baryon/meson ratios comparable to the bulk. No statistically significant differences were observed between Λ and $\bar{\Lambda}$ triggers or Ξ^- and Ξ^+ so these trigger particles were added together to increase the statistical significance of the results for comparisons to K_S^0 trigger particles. The $\Delta\phi$ was used because these measurements were statistically limited and the jet-like correlation is broader for identified particles than for unidentified particles.

Identified trigger particles

Figure 3.1 shows the $p_T^{trigger}$ dependence of Y_{Jet} for identified trigger particles in $d+Au$, $Cu + Cu$, and $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV for unidentified hadron, Λ , K_S^0 , and Ξ trigger particles. The $Au + Au$ data have not been corrected for the track merging effects discussed in Chapter 2, so the slightly lower yield for identified particles in $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV may be due to track merging. There is no system dependence within errors. The most central K_S^0 trigger in $Cu+Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV is about three sigma above the other data; this data point may be an outlier or may be an indication of a slight particle type dependence. There is no other observed dependence on the trigger particle type.

Figure 3.1: $p_T^{trigger}$ dependence of Y_{Jet} for $1.5 \text{ GeV}/c < p_T^{associated} < p_T^{trigger}$ in 0-60% central $Cu + Cu$ and 0-12% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200 \text{ GeV}$ for unidentified hadrons, Λ , K_S^0 , and Ξ trigger particles. The identified trigger particle data have been horizontally offset for visibility. The $Au + Au$ data have not been corrected for the track merging effects discussed in Chapter 2.

The $p_T^{associated}$ dependence of Y_{Jet} for unidentified hadrons, Λ , K_S^0 , and Ξ trigger particles is shown in Figure 3.2. Due to limited statistics, it was only possible to extract Y_{Jet} for two bins in $p_T^{associated}$ for Λ and K_S^0 trigger particles and one bin for Ξ trigger particles. The jet-like yields from identified trigger particles are within errors of the jet-like yields for unidentified hadron trigger particles. The data in $Au + Au$ collisions at $\sqrt{s_{NN}} = 200 \text{ GeV}$ also showed no dependence on trigger particle type [13].

The N_{part} dependence of Y_{Jet} is shown in Figure 3.3 for Λ , K_S^0 , and Ξ trigger particles. The $Au + Au$ data have not been corrected for track merging effects, and this may explain the difference between identified trigger particles and unidentified trigger particles.

No statistically significant differences between different trigger particles are observed for the jet-like correlations, including non-strange, singly strange, and doubly strange hadrons. Studies in $Au + Au$ collisions at $\sqrt{s_{NN}} = 200 \text{ GeV}$ indicated that there is no trigger particle dependence for even the trebly strange Ω [14, 15].

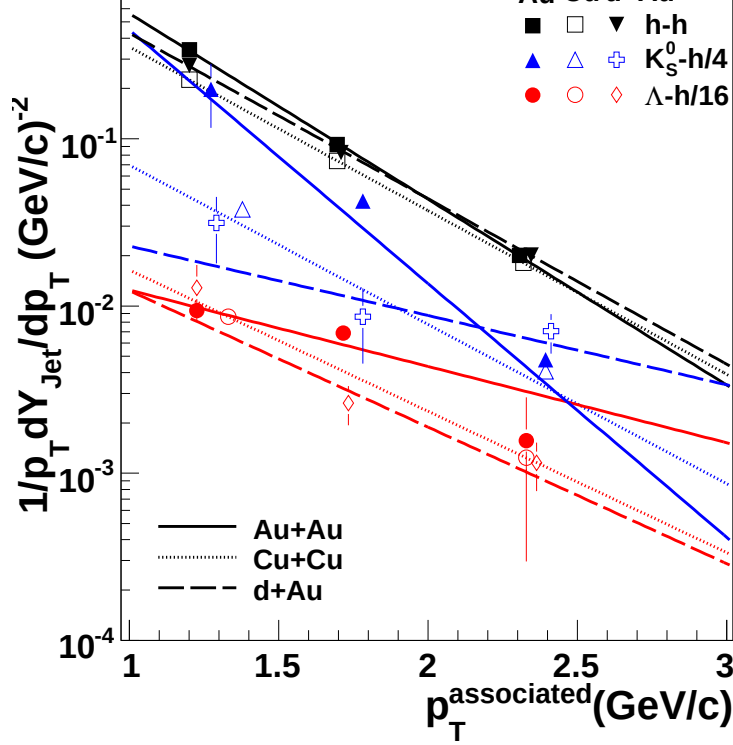


Figure 3.2: $p_T^{associated}$ dependence of Y_{Jet} for $3.0 < p_T^{trigger} < 6.0$ GeV/c in 0-60% central $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV for unidentified hadrons, Λ , K_S^0 , and Ξ trigger particles. The line is from a fit of the h-h data to $Ae^{-p_T/k}$. The inverse slope parameter is given in Table ???. The identified trigger particle data have been horizontally offset for visibility.

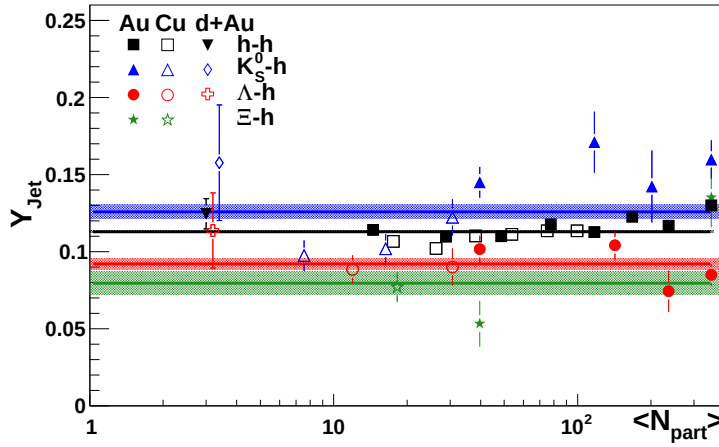


Figure 3.3: N_{part} dependence of Y_{Jet} for $3.0 < p_T^{trigger} < 6.0$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ for identified trigger particles in $d + Au$, $Cu + Cu$, and $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV for Λ , K_S^0 , and Ξ trigger particles.

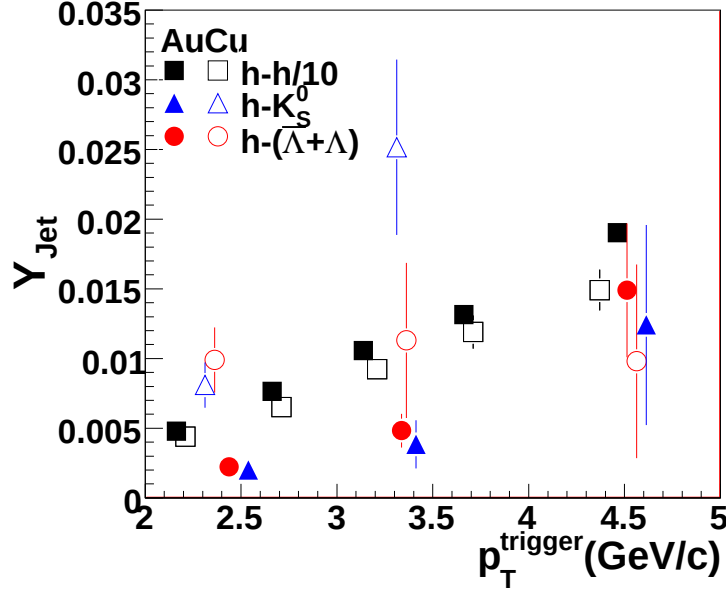


Figure 3.4: $p_T^{trigger}$ dependence of Y_{Jet} for $1.5 \text{ GeV/c} < p_T^{associated} < p_T^{trigger}$ of Y_{Jet} in 0-60% central $Cu + Cu$ and 0-12% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200 \text{ GeV}$ for identified associated particles. The unidentified hadron associated particles are scaled to fit on the same plot. Data from $Au + Au$ collisions at $\sqrt{s_{NN}} = 200 \text{ GeV}$ are from [13].

Table 3.1: Inverse slope parameter k (MeV/c) of $p_T^{associated}$ for fits of data from the jet-like correlation in Figure 3.5 to $Ae^{-p_T/k}$. Statistical errors only.

	h-h	h- K_S^0	h- $\Lambda + \bar{\Lambda}$
k (MeV/c)	445 ± 20	505 ± 87	510 ± 55

Identified associated particles

Figure 3.4 shows the $p_T^{trigger}$ dependence of Y_{Jet} for identified associated particles. Similar trends are observed for unidentified associated particles and strange associated particles.

Figure 3.5 shows the dependence of Y_{Jet} on $p_T^{associated}$ for identified associated particles. The ratio $\frac{\Lambda + \bar{\Lambda}}{2K_S^0}$ is roughly 0.5 for the full $p_T^{associated}$ range shown.

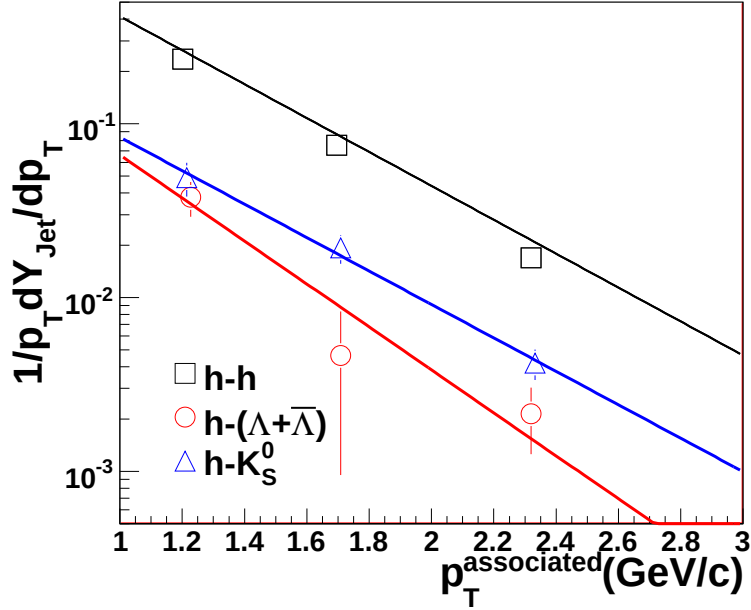


Figure 3.5: $p_T^{associated}$ dependence of Y_{Jet} for $3.0 < p_T^{trigger} < 6.0$ GeV/c for $\sqrt{s_{NN}} = 62$ GeV and $\sqrt{s_{NN}} = 200$ GeV. Parameters from fits of the data are listed in Table 3.1.

3.2 The *Ridge*

The *Ridge* is an unexpected phenomenon first observed in $Au + Au$ collisions. In order to understand its origins and to provide a robust test of models for the production of the *Ridge*, the *Ridge* is studied as a function of as many variables as possible. The collision energy, collision system, system size, and trigger particle type dependence of the *Ridge* are presented here.

3.2.1 Flavor dependence

Figure 3.6 shows the dependence of Y_{Ridge} on N_{part} for identified Λ , K_S^0 , and Ξ trigger particles for $Cu + Cu$ and $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV. The data for identified trigger particles in $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV are consistent, within large errors, of Y_{Ridge} in $Au + Au$ at the same N_{part} .

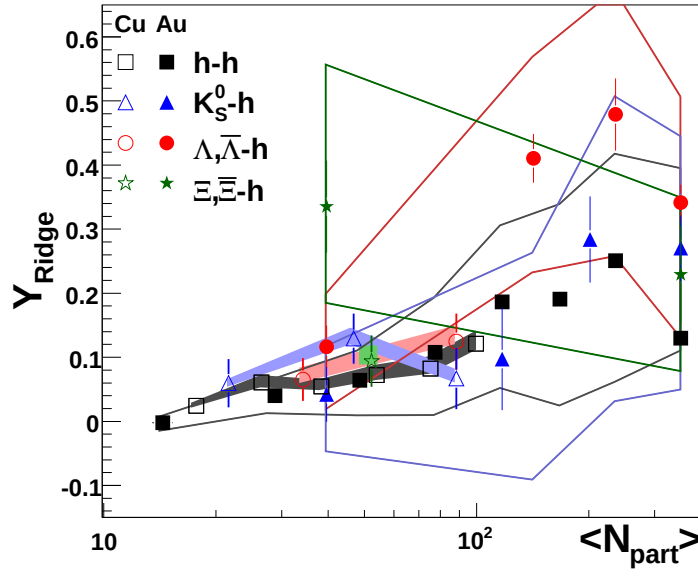


Figure 3.6: N_{part} dependence of Y_{Ridge} for $3.0 < p_T^{trigger} < 6.0$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ for identified trigger particles in $Cu+Cu$ and $Au+Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV. Systematic errors are shown as bands in the same color as the data points.

Appendix A

Geometric cuts

A.1 V^0 Geometric Cuts

Table A.1: Geometric cuts for Λ in $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV. Cuts for $\bar{\Lambda}$ the same with cuts for the p daughter of the Λ applied to the $\bar{p}$ daughter of the $\bar{\Lambda}$ and cuts for the π^- daughter of the Λ applied to the π^+ daughter of the $\bar{\Lambda}$. These are in addition to cuts listed in Table ???. Primary vertex is abbreviated to PV.

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
0.4 GeV/c < p_T < 0.6 GeV/c						
Decay length (cm) >	7.5	7.5	7.5	7.5	7.5	7.5
DCA to PV (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of p to PV (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA of π^- to PV (cm) >	2.3	2.3	2.3	1.5	1.5	0.0
$n_{\sigma, dE/dx}$ of daughters <	2.0	2.0	2.0	2.0	2.0	2.0
$ n_{\Gamma} \text{ from mass peak center} $	1.0	1.0	1.0	1.0	1.0	1.0
0.6 GeV/c < p_T < 0.8 GeV/c						
Decay length (cm) >	5.0	5.0	5.0	5.0	0.0	0.0
DCA to PV (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8

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Table A.1 – continued from previous page

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
DCA of p to PV (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA of π^- to PV (cm) >	2.2	2.2	2.2	1.5	1.5	1.5
$n_{\sigma,dE/dx}$ of daughters <	2.0	2.0	2.0	2.0	2.0	2.0
n_{Γ} from mass peak center i	2.0	2.0	2.0	2.0	2.0	2.0
0.8 GeV/c < p_T < 1.0 GeV/c						
Decay length (cm) >	6.5	0.0	0.0	0.0	0.0	0.0
DCA to PV (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of p to PV (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA of π^- to PV (cm) >	1.4	1.5	1.5	1.1	1.1	0.0
$n_{\sigma,dE/dx}$ of daughters <	2.0	2.0	2.0	2.0	2.0	2.0
n_{Γ} from mass peak center i	2.0	2.0	2.0	2.0	2.0	2.0
1.0 GeV/c < p_T < 1.5 GeV/c						
Decay length (cm) >	6.5	5.5	5.5	3.5	3.5	7.5
DCA to PV (cm) <	0.7	0.8	0.8	0.8	0.8	0.8
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of p to PV (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA of π^- to PV (cm) >	1.5	1.4	1.4	1.4	1.4	0.0
$n_{\sigma,dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
n_{Γ} from mass peak center i	2.0	2.0	2.0	2.0	2.0	2.0
1.5 GeV/c < p_T < 2.0 GeV/c						
Decay length (cm) >	9.0	7.0	7.0	0.0	0.0	6.0
DCA to PV (cm) <	0.7	0.7	0.8	0.8	0.8	0.8
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of p to PV (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA of π^- to PV (cm) >	1.4	1.5	1.5	1.4	1.4	0.0
$n_{\sigma,dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
n_{Γ} from mass peak center i	2.0	2.0	2.0	2.0	2.0	2.0
2.0 GeV/c < p_T < 2.5 GeV/c						
Decay length (cm) >	7.0	5.5	5.5	7.5	0.0	0.0

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Table A.1 – continued from previous page

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
DCA to PV (cm) <	0.7	0.4	0.4	0.8	0.8	0.8
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of p to PV (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA of π^- to PV (cm) >	1.7	1.4	1.4	1.4	1.4	1.6
$n_{\sigma,dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
n_Γ from mass peak center i	2.0	2.0	2.0	2.0	2.0	2.0
2.5 GeV/c < p_T < 3.0 GeV/c						
Decay length (cm) >	15.0	11.0	12.0	11.0	12.0	0.0
DCA to PV (cm) <	0.5	0.8	0.8	0.8	0.8	0.8
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of p to PV (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA of π^- to PV (cm) >	1.4	1.2	1.8	1.0	1.8	1.5
$n_{\sigma,dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
n_Γ from mass peak center i	2.0	2.0	2.0	2.0	2.0	2.0
3.0 GeV/c < p_T < 3.5 GeV/c						
Decay length (cm) >	17.0	8.0	8.0	8.0	8.0	8.0
DCA to PV (cm) <	0.5	0.5	0.5	0.5	0.5	0.5
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of p to PV (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA of π^- to PV (cm) >	1.7	1.6	1.6	1.2	1.2	0.6
$n_{\sigma,dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
n_Γ from mass peak center i	2.0	2.0	2.0	2.0	2.0	2.0
3.5 GeV/c < p_T < 4.0 GeV/c						
Decay length (cm) >	13.0	13.0	13.0	13.0	13.0	13.0
DCA to PV (cm) <	0.4	0.4	0.4	0.4	0.4	0.4
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of p to PV (cm) >	0.3	0.3	0.3	0.3	0.3	0.3
DCA of π^- to PV (cm) >	1.2	1.2	1.2	1.2	1.2	1.2
$n_{\sigma,dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
n_Γ from mass peak center i	1.5	1.5	1.5	1.5	1.5	1.5

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Table A.1 – continued from previous page

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
4.0 GeV/c < p_T < 5.0 GeV/c						
Decay length (cm) >	15.0	15.0	15.0	15.0	15.0	15.0
DCA to PV (cm) <	0.4	0.4	0.4	0.4	0.4	0.4
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of p to PV (cm) >	0.3	0.3	0.3	0.3	0.3	0.3
DCA of π^- to PV (cm) >	1.2	1.2	1.2	1.2	1.2	1.2
$n_{\sigma, dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
n_Γ from mass peak center i	1.5	1.5	1.5	1.5	1.5	1.5
5.0 GeV/c < p_T < 10.0 GeV/c						
Decay length (cm) >	7.5	7.5	7.5	7.5	0.0	7.5
DCA to PV (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of p to PV (cm) >	1.5	1.5	1.5	1.5	1.5	3.0
DCA of π^- to PV (cm) >	1.5	1.5	1.5	1.5	0.0	1.5
$n_{\sigma, dE/dx}$ of daughters <	2.0	2.0	2.0	2.0	2.0	2.0
n_Γ from mass peak center i	1.5	1.5	1.5	1.5	1.5	1.5

Table A.2: Geometric cuts for K_S^0 in $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV. These are in addition to cuts listed in Table ?? . Primary vertex is abbreviated to PV.

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
0.4 GeV/c < p_T < 0.6 GeV/c						
Decay length (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA to PV (cm) <	0.6	0.8	0.8	0.8	0.8	0.8
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of π^- , π^+ to PV (cm) >	2.1	1.1	1.1	1.1	1.1	1.1
$n_{\sigma, dE/dx}$ of daughters <	2.0	2.0	2.0	2.0	2.0	2.0
n_Γ from mass peak center i	1.5	1.5	1.5	1.5	1.5	1.5

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Table A.2 – continued from previous page

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
0.6 GeV/c < p_T < 0.8 GeV/c						
Decay length (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA to PV (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of π^- , π^+ to PV (cm) >	1.0	1.0	1.0	1.0	1.0	1.0
$n_{\sigma, dE/dx}$ of daughters <	2.0	2.0	2.0	2.0	2.0	2.0
n_Γ from mass peak center i	1.5	1.5	1.5	1.5	2.0	2.0
0.8 GeV/c < p_T < 1.0 GeV/c						
Decay length (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA to PV (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of π^- , π^+ to PV (cm) >	0.9	0.9	0.9	0.9	0.9	0.9
$n_{\sigma, dE/dx}$ of daughters <	2.0	2.0	2.0	2.0	2.0	2.0
n_Γ from mass peak center i	1.5	1.5	1.5	2.0	2.0	2.0
1.0 GeV/c < p_T < 1.5 GeV/c						
Decay length (cm) >	4.5	4.5	4.5	4.5	4.5	4.5
DCA to PV (cm) <	0.6	0.6	0.6	0.8	0.8	0.8
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of π^- , π^+ to PV (cm) >	0.9	0.9	0.9	0.9	0.9	0.9
$n_{\sigma, dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
n_Γ from mass peak center i	1.5	1.5	1.5	1.5	1.5	2.0
1.5 GeV/c < p_T < 2.0 GeV/c						
Decay length (cm) >	4.5	4.5	4.5	4.5	4.5	4.5
DCA to PV (cm) <	0.4	0.4	0.4	0.8	0.8	0.8
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of π^- , π^+ to PV (cm) >	1.5	1.5	1.5	1.2	1.1	0.0
$n_{\sigma, dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
n_Γ from mass peak center i	1.5	2.0	2.0	1.5	1.5	1.5
2.0 GeV/c < p_T < 2.5 GeV/c						
Decay length (cm) >	0.0	0.0	0.0	0.0	0.0	0.0

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Table A.2 – continued from previous page

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
DCA to PV (cm) <	0.4	0.4	0.4	0.8	0.8	0.8
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of π^- , π^+ to PV (cm) >	1.6	1.4	1.4	1.4	1.1	1.1
$n_{\sigma,dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
n_{Γ} from mass peak center i	1.5	2.0	2.0	1.5	1.5	1.5
2.5 GeV/c < p_T < 3.0 GeV/c						
Decay length (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA to PV (cm) <	0.4	0.4	0.4	0.4	0.4	0.4
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of π^- , π^+ to PV (cm) >	1.2	1.2	1.0	0.0	0.0	0.0
$n_{\sigma,dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
n_{Γ} from mass peak center i	1.5	2.0	2.0	1.5	2.0	2.0
3.0 GeV/c < p_T < 3.5 GeV/c						
Decay length (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA to PV (cm) <	0.4	0.4	0.4	0.5	0.5	0.5
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of π^- , π^+ to PV (cm) >	0.8	0.8	0.8	0.8	0.8	0.8
$n_{\sigma,dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
n_{Γ} from mass peak center i	1.5	1.5	1.5	1.5	1.5	2.0
3.5 GeV/c < p_T < 4.0 GeV/c						
Decay length (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA to PV (cm) <	0.4	0.4	0.4	0.4	0.4	0.4
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of π^- , π^+ to PV (cm) >	0.8	0.8	0.6	0.6	0.6	0.6
$n_{\sigma,dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
n_{Γ} from mass peak center i	1.5	1.5	1.5	1.5	1.5	1.5
4.0 GeV/c < p_T < 5.0 GeV/c						
Decay length (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA to PV (cm) <	0.4	0.4	0.4	0.4	0.4	0.4
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8

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Table A.2 – continued from previous page

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
DCA of π^- , π^+ to PV (cm) >	0.8	0.8	0.6	0.6	0.6	0.6
$n_{\sigma, dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
n_{Γ} from mass peak center i	2.0	2.0	2.0	2.0	2.0	2.0
5.0 GeV/c < p_T < 10.0 GeV/c						
Decay length (cm) >	0.0	0.0	0.0	0.0	0.0	0.2
DCA to PV (cm) <	0.4	0.4	0.6	0.6	0.8	1.5
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	2.0
DCA of π^- , π^+ to PV (cm) >	1.4	1.4	1.4	1.4	1.4	1.5
$n_{\sigma, dE/dx}$ of daughters <	1.5	1.5	2.0	2.0	2.0	7.5
n_{Γ} from mass peak center i	2.0	2.0	2.0	2.0	2.0	0.0

A.2 Ξ^- Geometric Cuts

Table A.3: Geometric cuts for Ξ^- in $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV. These are in addition to cuts listed in Table ???. Cuts for Ξ^+ the same with cuts for the p daughter of the Λ applied to the $\bar{p}$ daughter of the $\bar{\Lambda}$, cuts for the π^- daughter of the Λ applied to the π^+ daughter of the $\bar{\Lambda}$, and cuts for the Λ applied to the $\bar{\Lambda}$. A and B refer to the parameters in Equation ??. Primary vertex is abbreviated to PV.

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
0.6 GeV/c < p_T < 0.8 GeV/c						
Ξ^- Decay length (cm)	> 3.5	> 3.5	> 3.5	> 3.5	> 3.5	> 3.5
Λ Decay length (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of Ξ^- to primary vertex (cm)	< 0.55	< 0.55	< 0.55	< 0.55	< 0.55	< 0.55
DCA of Λ to primary vertex (cm)	> 0.18	> 0.18	> 0.18	> 0.18	> 0.18	> 0.18
DCA of Ξ^- to daughters (cm)	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6
DCA of Λ to daughters (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of p to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0

Continued on next page

Table A.3 – continued from previous page

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
DCA of π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of bachelor π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
$n_{\sigma, dE/dx}$ of daughters	> 3	> 3	> 3	> 3	> 3	> 3
n_Γ from mass peak center	> 1.5	> 1.5	> 2	> 2	> 2	> 2
A	1.8333	1.8333	1.8333	1.8333	1.3333	1.8333
B	-0.03	-0.03	-0.03	-0.05	-0.37	-0.06
0.8 GeV/c < p_T < 1 GeV/c						
Ξ^- Decay length (cm)	> 3.5	> 3.5	> 3.5	> 3.5	> 3.5	> 3.5
Λ Decay length (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of Ξ^- to primary vertex (cm)	< 0.55	< 0.55	< 0.55	< 0.55	< 0.55	< 0.55
DCA of Λ to primary vertex (cm)	> 0.18	> 0.18	> 0.18	> 0.18	> 0.18	> 0.18
DCA of Ξ^- to daughters (cm)	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6
DCA of Λ to daughters (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of p to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of bachelor π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
$n_{\sigma, dE/dx}$ of daughters	> 3	> 3	> 3	> 3	> 3	> 3
n_Γ from mass peak center	> 1.5	> 1.5	> 2	> 2	> 2	> 2
A	1.8333	1.8333	1.8333	1.3333	N/A	1.8333
B	-0.2	-0.2	-0.2	-1000	N/A	-0.06
1 GeV/c < p_T < 1.5 GeV/c						
Ξ^- Decay length (cm)	> 3.5	> 3.5	> 3.5	> 3.5	> 3.5	> 3.5
Λ Decay length (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of Ξ^- to primary vertex (cm)	< 0.55	< 0.55	< 0.55	< 0.55	< 0.55	< 0.55
DCA of Λ to primary vertex (cm)	> 0.18	> 0.18	> 0.18	> 0.18	> 0.18	> 0.18
DCA of Ξ^- to daughters (cm)	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6
DCA of Λ to daughters (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of p to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of bachelor π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0

Continued on next page

Table A.3 – continued from previous page

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
$n_{\sigma, dE/dx}$ of daughters	> 3	> 3	> 3	> 3	> 3	> 3
n_{Γ} from mass peak center	> 1.5	> 1.5	> 2	> 2	> 2	> 2
A	1.3333	1.3333	1.3333	N/A	N/A	1.8333
B	-0.35	-0.35	-0.35	N/A	N/A	-0.06
1.5 GeV/c < p_T < 2 GeV/c						
Ξ^- Decay length (cm)	> 3.5	> 3.5	> 3.5	> 3.5	> 3.5	> 3.5
Λ Decay length (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of Ξ^- to primary vertex (cm)	< 0.55	< 0.55	< 0.55	< 0.55	< 0.55	< 0.55
DCA of Λ to primary vertex (cm)	> 0.18	> 0.18	> 0.18	> 0.18	> 0.18	> 0.18
DCA of Ξ^- to daughters (cm)	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6
DCA of Λ to daughters (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of p to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of bachelor π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
$n_{\sigma, dE/dx}$ of daughters	> 3	> 3	> 3	> 3	> 3	> 3
n_{Γ} from mass peak center	> 1.5	> 1.5	> 2	> 2	> 2	> 2
A	1.3333	1.3333	1.3333	N/A	N/A	1.8333
B	-0.4	-0.4	-0.4	N/A	N/A	-0.06
2 GeV/c < p_T < 2.5 GeV/c						
Ξ^- Decay length (cm)	> 3.5	> 3.5	> 3.5	> 3.5	> 3.5	> 3.5
Λ Decay length (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of Ξ^- to primary vertex (cm)	< 0.55	< 0.55	< 0.55	< 0.55	< 0.55	< 0.55
DCA of Λ to primary vertex (cm)	> 0.18	> 0.18	> 0.18	> 0.18	> 0.18	> 0.18
DCA of Ξ^- to daughters (cm)	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6
DCA of Λ to daughters (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of p to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of bachelor π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
$n_{\sigma, dE/dx}$ of daughters	> 3	> 3	> 3	> 3	> 3	> 3
n_{Γ} from mass peak center	> 1.5	> 1.5	> 2	> 2	> 2	> 2

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Table A.3 – continued from previous page

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
A	1.8333	1.8333	1.8333	1.3333	1.8333	1.8333
B	0.05	0.05	0.05	0	-0.06	0
2.5 GeV/c < p_T < 3 GeV/c						
Ξ^- Decay length (cm)	> 3	> 3	> 3	> 3	> 3	> 3
Λ Decay length (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of Ξ^- to primary vertex (cm)	< 0.45	< 0.45	< 0.45	< 0.45	< 0.45	< 0.45
DCA of Λ to primary vertex (cm)	> 0.1	> 0.1	> 0.1	> 0.1	> 0.1	> 0.1
DCA of Ξ^- to daughters (cm)	< 0.45	< 0.45	< 0.45	< 0.45	< 0.45	< 0.45
DCA of Λ to daughters (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of p to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of bachelor π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
$n_{\sigma, dE/dx}$ of daughters	> 3	> 3	> 3	> 3	> 3	> 3
n_Γ from mass peak center	> 1.5	> 1.5	> 2	> 2	> 2	> 2
A	1.8333	1.8333	1.8333	1.3333	1.8333	1.8333
B	0	0	0.05	0	-0.06	0
3 GeV/c < p_T < 3.5 GeV/c						
Ξ^- Decay length (cm)	> 0	> 0	> 0	> 0	> 0	> 0
Λ Decay length (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of Ξ^- to primary vertex (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of Λ to primary vertex (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of Ξ^- to daughters (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of Λ to daughters (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of p to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of bachelor π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
$n_{\sigma, dE/dx}$ of daughters	> 3	> 3	> 3	> 3	> 3	> 3
n_Γ from mass peak center	> 2	> 2	> 2	> 2	> 2	> 2
A	1.8333	1.8333	1.8333	1.3333	1.8333	1.8333
B	-0.03	-0.03	-0.05	-0.37	-0.06	0

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Table A.3 – continued from previous page

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
3.5 GeV/c < p_T < 4.5 GeV/c						
Ξ^- Decay length (cm)	> 0	> 0	> 0	> 0	> 0	> 0
Λ Decay length (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of Ξ^- to primary vertex (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of Λ to primary vertex (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of Ξ^- to daughters (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of Λ to daughters (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of p to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of bachelor π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
$n_{\sigma, dE/dx}$ of daughters	> 3	> 3	> 3	> 3	> 3	> 3
n_Γ from mass peak center	> 2	> 2	> 2	> 2	> 2	> 2
A	1.8333	1.8333	1.3333	N/A	1.8333	N/A
B	-0.2	-0.2	-1000	N/A	-0.06	N/A
$p_T > 4.5$ GeV/c						
Ξ^- Decay length (cm)	> 0	> 0	> 0	> 0	> 0	> 0
Λ Decay length (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of Ξ^- to primary vertex (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of Λ to primary vertex (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of Ξ^- to daughters (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of Λ to daughters (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of p to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of bachelor π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
$n_{\sigma, dE/dx}$ of daughters	> 3	> 3	> 3	> 3	> 3	> 3
n_Γ from mass peak center	> 2	> 2	> 2	> 2	> 2	> 2
A	1.3333	1.3333	N/A	N/A	1.8333	N/A
B	-0.35	-0.35	N/A	N/A	-0.06	N/A

Appendix B

Efficiencies

Table B.1: Fit parameters from fits of the efficiencies for unidentified hadrons in $Au + Au$ at $\sqrt{s_{NN}} = 62$ GeV to $a \exp(\frac{b}{p_T})^c$

centrality	a	b	c
0-5%	.767	.1498	1.316
5-10%	.7856	.1419	1.33
10-20%	.7975	.1408	1.464
20-30%	.8113	.1374	1.606
30-40%	.822	.1376	1.743
40-50%	.8321	.1327	1.781
50-60%	.8386	.1357	1.912
60-70%	.8387	.1511	2.381
70-80%	.8415	.1455	2.153

Table B.2: Fit parameters from fits of the efficiencies for unidentified hadrons in $Au + Au$ at $\sqrt{s_{NN}} = 62$ GeV to $a \exp(\frac{b}{p_T})^c$

centrality	a	b	c
0-10%	0.838435	0.0704	0.9409
10-20%	0.843666	0.0681	1.0000
20-30%	0.843211	0.0610	1.0000
30-40%	0.848090	0.0638	1.0000
40-50%	0.856875	0.0606	0.9496
50-60%	0.851112	0.0584	1.0000

Table B.3: Fit parameters from fits of the efficiencies for unidentified hadrons in $Au + Au$ at $\sqrt{s_{NN}} = 62$ GeV to $a \exp(\frac{b}{p_T})^c$

centrality	a	b	c
0-10%	0.8655	0.1333	0.9890
10-20%	0.8736	0.1207	0.9330
20-30%	0.8643	0.1503	1.1071
30-40%	0.8693	0.1269	0.9922
40-50%	0.8608	0.1511	1.1616
50-60%	0.8606	0.1971	1.3211

Table B.4: Fit parameters from fits of the efficiencies for unidentified hadrons in $Au + Au$ at $\sqrt{s_{NN}} = 200$ GeV to $a \exp(\frac{b}{p_T})^c$

centrality	a	b	c
0-5%	.767	.1498	1.316
5-10%	.7856	.1419	1.33
10-20%	.7975	.1408	1.464
20-30%	.8113	.1374	1.606
30-40%	.822	.1376	1.743
40-50%	.8321	.1327	1.781
50-60%	.8386	.1357	1.912
60-70%	.8387	.1511	2.381
70-80%	.8415	.1455	2.153

B.1 V^0 Efficiencies

B.1.1 V^0 Efficiencies in $Cu+Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV

Table B.5: Fit parameters from fits of the efficiencies for Λ baryons in $Cu + Cu$ at $\sqrt{s_{NN}} = 200$ GeV to $a \exp(\frac{b}{p_T})^c$

		0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
$1 < p_T < 1.5$	a	0.172	0.164	0.168	0.158	0.161	0.166
	b	0.946	0.838	0.819	0.774	0.758	0.861
	c	1.587	1.800	1.825	1.902	2.020	1.895
$1.5 < p_T < 2$	a	0.174	0.162	0.169	0.159	0.161	0.164
	b	1.043	0.887	0.855	0.772	0.755	0.821
	c	1.701	1.884	1.835	1.842	2.033	1.947
$2 < p_T < 2.5$	a	0.172	0.145	0.152	0.159	0.161	0.163
	b	0.969	0.915	0.913	0.845	0.755	0.789
	c	1.586	2.106	2.066	1.999	2.033	1.790
$2.5 < p_T < 3$	a	0.177	0.152	0.157	0.149	0.154	0.159
	b	1.468	0.953	0.924	0.889	0.911	0.926
	c	1.890	2.120	2.149	2.280	2.406	2.023
$3 < p_T < 3.5$	a	0.177	0.152	0.157	0.149	0.154	0.159
	b	1.468	0.953	0.924	0.889	0.911	0.926
	c	1.890	2.120	2.149	2.280	2.406	2.023
$3.5 < p_T < 4$	a	0.137	0.110	0.198	0.246	0.252	0.140
	b	0.038	0.051	0.023	0.011	0.009	0.048
	c	-0.005	-0.006	-0.004	-0.003	-0.003	-0.006
$4 < p_T < 5$	a	0.137	0.110	0.198	0.246	0.252	0.140
	b	0.038	0.051	0.023	0.011	0.009	0.048
	c	-0.005	-0.006	-0.004	-0.003	-0.003	-0.006
$p_T > 5$	a	0.177	0.152	0.157	0.149	0.154	0.159
	b	1.468	0.953	0.924	0.889	0.911	0.926
	c	1.890	2.120	2.149	2.280	2.406	2.023

Table B.6: Fit parameters from fits of the efficiencies for $\bar{\Lambda}$ baryons in $Au + Au$ at $\sqrt{s_{NN}} = 62$ GeV to $a \exp(\frac{b}{p_T})^c$

		0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
$1 < p_T < 1.5$	a	0.158	0.163	0.169	0.169	0.186	0.196
	b	0.924	0.888	0.884	0.784	0.897	1.015
	c	1.767	1.683	1.752	1.659	1.479	1.444
$1.5 < p_T < 2$	a	0.163	0.158	0.174	0.178	0.187	0.205
	b	1.092	1.054	1.076	1.026	1.104	1.200
	c	1.948	2.156	1.950	1.763	1.758	1.527
$2 < p_T < 2.5$	a	0.161	0.139	0.152	0.167	0.189	0.198
	b	1.042	0.946	0.968	0.860	0.904	0.981
	c	1.893	2.284	2.084	1.916	1.402	1.294
$2.5 < p_T < 3$	a	0.163	0.146	0.155	0.155	0.165	0.183
	b	1.414	0.967	0.953	0.914	1.006	1.058
	c	1.983	2.281	2.155	2.206	2.166	1.632
$3 < p_T < 3.5$	a	0.163	0.146	0.155	0.155	0.165	0.183
	b	1.414	0.967	0.953	0.914	1.006	1.058
	c	1.983	2.281	2.155	2.206	2.166	1.632
$3.5 < p_T < 4$	a	0.113	0.149	0.227	0.148	0.072	0.224
	b	0.047	0.036	0.010	0.042	0.070	0.013
	c	-0.006	-0.005	-0.002	-0.005	-0.008	-0.003
$4 < p_T < 5$	a	0.113	0.149	0.227	0.148	0.072	0.224
	b	0.047	0.036	0.010	0.042	0.070	0.013
	c	-0.006	-0.005	-0.002	-0.005	-0.008	-0.003
$p_T > 5$	a	0.163	0.146	0.155	0.155	0.165	0.183
	b	1.414	0.967	0.953	0.914	1.006	1.058
	c	1.983	2.281	2.155	2.206	2.166	1.632

Table B.7: Fit parameters from fits of the efficiencies for K_S^0 baryons in $Au + Au$ at $\sqrt{s_{NN}} = 62$ GeV to $a \exp(\frac{b}{p_T})^c$

		0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
$1 < p_T < 1.5$	a	0.176	0.174	0.280	0.270	0.257	0.294
	b	0.972	0.917	1.065	1.014	0.964	1.129
	c	1.606	1.796	1.589	1.778	1.924	1.563
$1.5 < p_T < 2$	a	0.176	0.174	0.280	0.270	0.257	0.299
	b	0.972	0.917	1.065	1.014	0.964	1.000
	c	1.606	1.796	1.589	1.778	1.924	1.298
$2 < p_T < 2.5$	a	0.194	0.182	0.284	0.275	0.262	0.299
	b	0.989	0.895	0.936	0.907	0.866	1.000
	c	1.222	1.502	1.329	1.511	1.652	1.298
$2.5 < p_T < 3$	a	0.194	0.182	0.284	0.275	0.262	0.299
	b	0.989	0.895	0.936	0.907	0.866	1.000
	c	1.222	1.502	1.329	1.511	1.652	1.298
$3 < p_T < 3.5$	a	0.194	0.182	0.284	0.275	0.262	0.299
	b	0.989	0.895	0.936	0.907	0.866	1.000
	c	1.222	1.502	1.329	1.511	1.652	1.298
$3.5 < p_T < 4$	a	0.127	0.114	0.164	0.141	0.177	0.234
	b	0.014	0.022	0.041	0.051	0.038	0.014
	c	-0.002	-0.003	-0.004	-0.005	-0.004	-0.002
$4 < p_T < 5$	a	0.127	0.114	0.164	0.141	0.177	0.234
	b	0.014	0.022	0.041	0.051	0.038	0.014
	c	-0.002	-0.003	-0.004	-0.005	-0.004	-0.002
$p_T > 5$	a	0.194	0.182	0.284	0.275	0.262	0.299
	b	0.989	0.895	0.936	0.907	0.866	1.000
	c	1.222	1.502	1.329	1.511	1.652	1.298

B.1.2 V^0 Efficiencies in $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV

Table B.8: Fit parameters from fits of the efficiencies for Λ in $Au + Au$ at $\sqrt{s_{NN}} = 200$ GeV to $a \exp(\frac{b}{p_T}) + d + e$

centrality	a	b	c	d	e
0-5%	9.73455e-02	1.44024e+00	1.17638e+00	-9.13192e-04	-8.74599e-04
5-10%	1.05882e-01	1.41282e+00	1.23304e+00	-4.41302e-04	-1.03846e-03
10-20%	9.98811e-02	1.21169e+00	1.41522e+00	-1.42956e-04	-9.95265e-04
20-30%	1.23769e-01	1.37160e+00	1.17591e+00	-5.06965e-04	-1.17704e-03
30-40%	1.09135e-01	1.20635e+00	1.34792e+00	1.98552e-03	-1.37347e-03
40-50%	1.07304e-01	1.16138e+00	1.54782e+00	2.08468e-03	-1.35536e-03
50-60%	8.70185e-02	1.02422e+00	2.14817e+00	3.93120e-03	-1.29337e-03
60-70%	7.30531e-02	1.01570e+00	2.43480e+00	8.27065e-03	-1.70979e-03
70-80%	9.63416e-02	1.24434e+00	1.25699e+00	5.21417e-03	-1.63667e-03

Table B.9: Fit parameters from fits of the efficiencies for $\bar{\Lambda}$ in $Au + Au$ at $\sqrt{s_{NN}} = 200$ GeV to $a \exp(\frac{b}{p_T}) + d + e$

centrality	a	b	c	d	e
0-5%	9.11053e-02	1.38980e+00	1.24573e+00	3.35935e-04	-9.73930e-04
5-10%	1.00155e-01	1.29565e+00	1.28446e+00	1.75720e-04	-1.06484e-03
10-20%	9.38267e-02	1.22126e+00	1.48764e+00	1.83101e-03	-1.18936e-03
20-30%	9.90359e-02	1.14356e+00	1.48145e+00	9.35607e-04	-1.07923e-03
30-40%	1.01358e-01	1.11225e+00	1.52444e+00	2.63142e-03	-1.36184e-03
40-50%	1.07008e-01	1.11577e+00	1.50770e+00	2.32430e-03	-1.39252e-03
50-60%	9.11440e-02	1.06653e+00	1.91641e+00	4.72654e-03	-1.44842e-03
60-70%	9.09852e-02	1.09978e+00	1.60967e+00	5.31837e-03	-1.54718e-03
70-80%	9.74573e-02	9.86275e-01	1.77140e+00	-2.77767e-03	-6.33058e-04

Table B.10: Fit parameters from fits of the efficiencies for K_S^0 in $Au + Au$ at $\sqrt{s_{NN}} = 200$ GeV to $a \exp(\frac{b}{p_T}) + d + e$

centrality	a	b	c	d	e
0-5%	1.90006e-01	1.46579e+00	1.58672e+00	1.18775e-03	-1.16531e-03
5-10%	1.99357e-01	1.42488e+00	1.62480e+00	2.00574e-03	-1.21733e-03
10-20%	2.11215e-01	1.36336e+00	1.65828e+00	1.71577e-03	-1.28160e-03
20-30%	2.25984e-01	1.35987e+00	1.60081e+00	1.44734e-03	-1.37115e-03
30-40%	2.20745e-01	1.26539e+00	1.75534e+00	2.46239e-03	-1.38532e-03
40-50%	2.01929e-01	1.19933e+00	2.04934e+00	5.35690e-03	-1.44794e-03
50-60%	2.34387e-01	1.25719e+00	1.68057e+00	2.42561e-03	-1.56188e-03
60-70%	1.94564e-01	1.23620e+00	2.26538e+00	5.01216e-03	-1.36288e-03
70-80%	1.80066e-01	1.14058e+00	2.16036e+00	2.52093e-03	-8.10902e-04

Appendix C

Centrality

	R	N_{part}	N_{bin}
0-10%	$101 \leq R$	96	162
10-20%	$71 \leq R < 101$	72	108
20-30%	$49 \leq R < 71$	52	68
30-40%	$33 \leq R < 49$	36	42
40-50%	$22 \leq R < 33$	25	26
50-60%	$14 \leq R < 22$	16	15

Table C.1: Centrality definitions for $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 62$ GeV

	R	N_{part}	N_{bin}
0-10%	$139 \leq R$	99	189
10-20%	$98 \leq R < 139$	75	124
20-30%	$67 \leq R < 98$	54	78
30-40%	$46 \leq R < 67$	38	48
40-50%	$30 \leq R < 46$	26	29
50-60%	$19 \leq R < 30$	17	17

Table C.2: Centrality definitions for $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV

	R	N_{part}	N_{bin}
0-5%	$373 \geq R$	347	904
5-10%	$313 \leq R < 373$	293	714
10-20%	$222 \leq R < 313$	229	512
20-30%	$154 \leq R < 222$	162	321
30-40%	$102 \leq R < 154$	112	193
40-50%	$65 \leq R < 102$	74	109
50-60%	$38 \leq R < 65$	46	57
60-70%	$20 \leq R < 38$	26	27
70-80%	$9 \leq R < 20$	13	11

Table C.3: Centrality definitions for $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV

	R	N_{part}	N_{bin}
0-5%	$510 \geq R$	352.4	1051.3
5-10%	$431 \leq R < 510$	299.3	827.9
10-20%	$312 \leq R < 431$	234.6	591.3
20-30%	$217 \leq R < 312$	166.7	368.6
30-40%	$146 \leq R < 217$	115.5	220.2
40-50%	$94 \leq R < 146$	76.6	123.4
50-60%	$56 \leq R < 94$	47.8	63.9
60-70%	$30 \leq R < 56$	27.4	29.5
70-80%	$14 \leq R < 30$	14.1	12.3

Table C.4: Centrality definitions for $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV

Appendix D

$p_T^{trigger}$ dependence for different collision energies

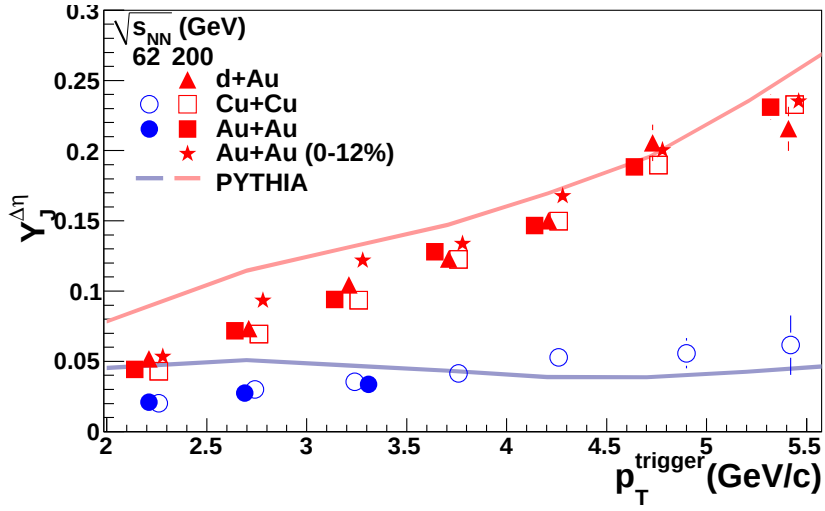


Figure D.1: $p_T^{trigger}$ dependence of Y_{Jet} for unidentified hadrons for $1.5 \text{ GeV/c} < p_T^{associated} < p_T^{trigger}$ for 0-60% central $Cu + Cu$ and 0-80% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV and minimum bias $d + Au$, 0-60% central $Cu + Cu$ and 40-80% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

All points are done using bin counting and the $\Delta\eta$ method.

Appendix E

Fits for $p_T^{trigger}$ dependence in $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 62$ GeV

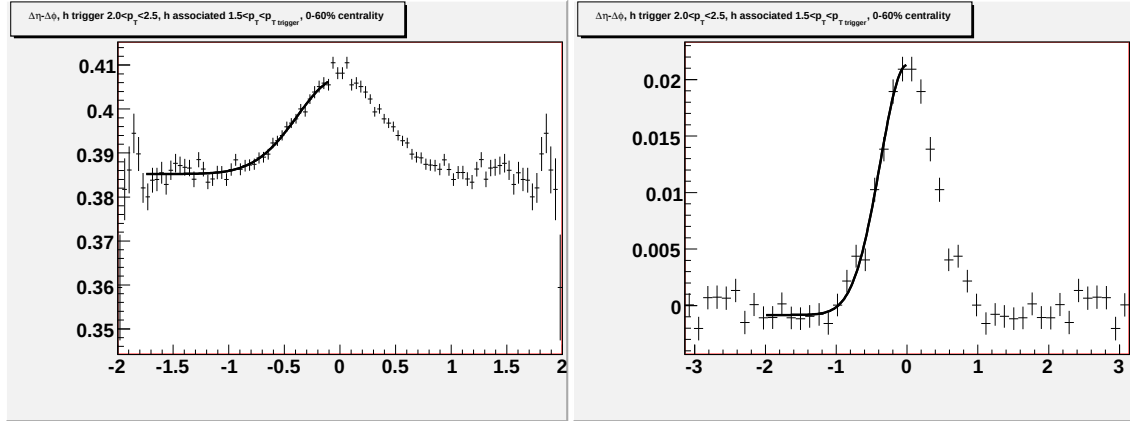


Figure E.1: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $2.0 < p_T^{trigger} < 2.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

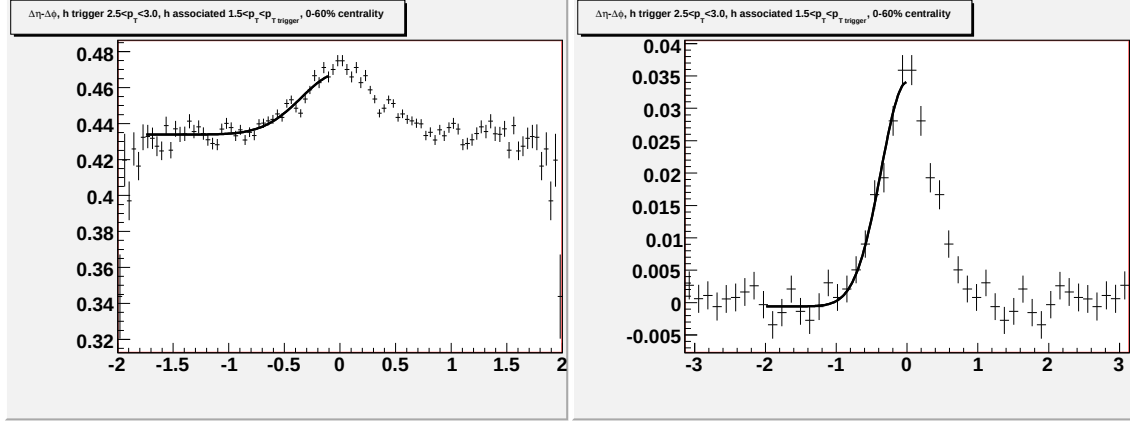


Figure E.2:]

Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $2.5 < p_T^{trigger} < 3.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

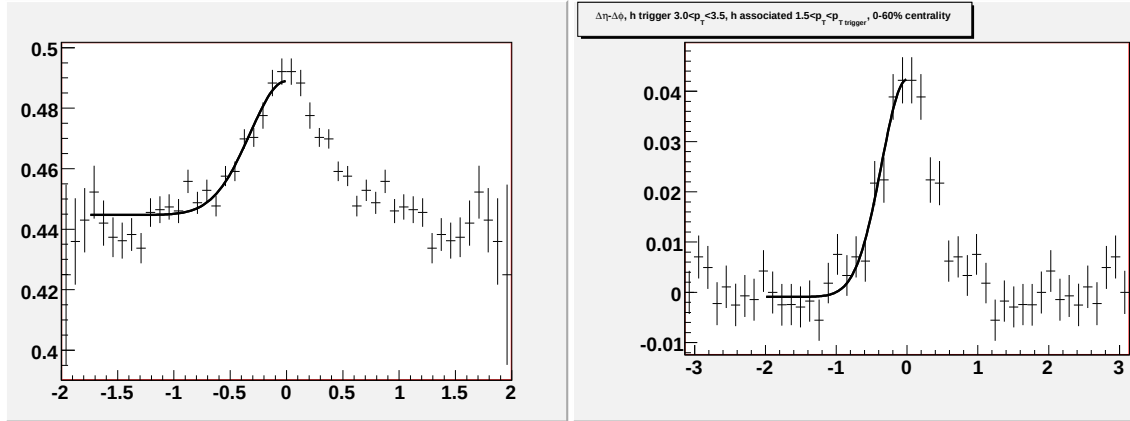


Figure E.3:]

Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 3.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

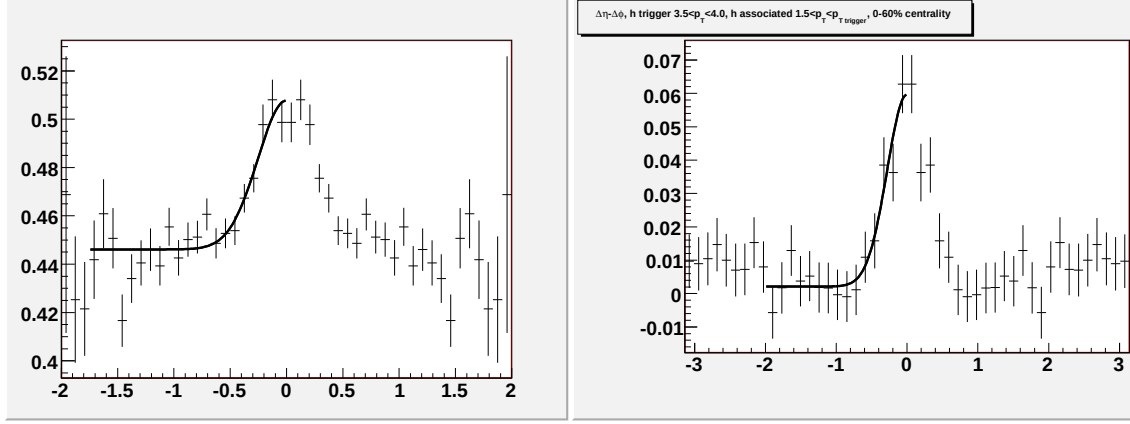


Figure E.4:]

Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.5 < p_T^{trigger} < 4.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

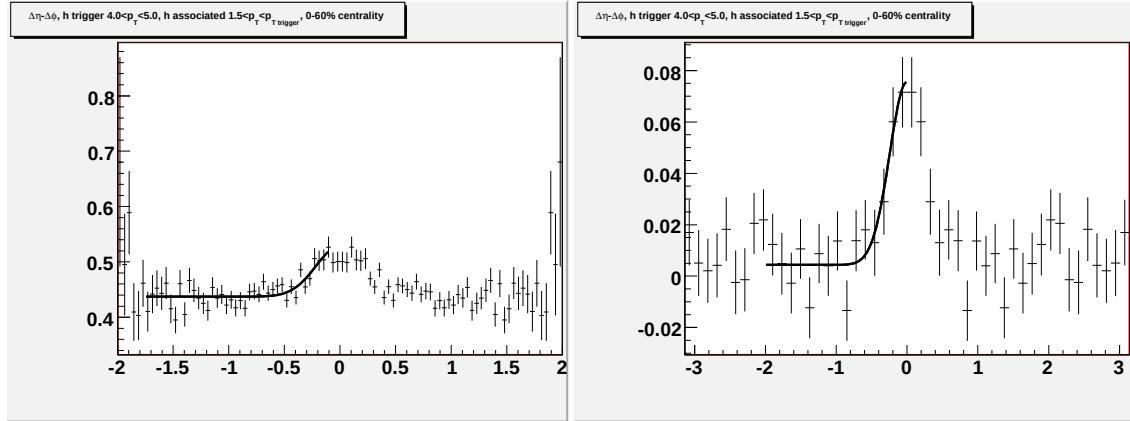


Figure E.5:]

Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $4.0 < p_T^{trigger} < 5.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

Appendix F

Fits for $p_T^{trigger}$ dependence in $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV

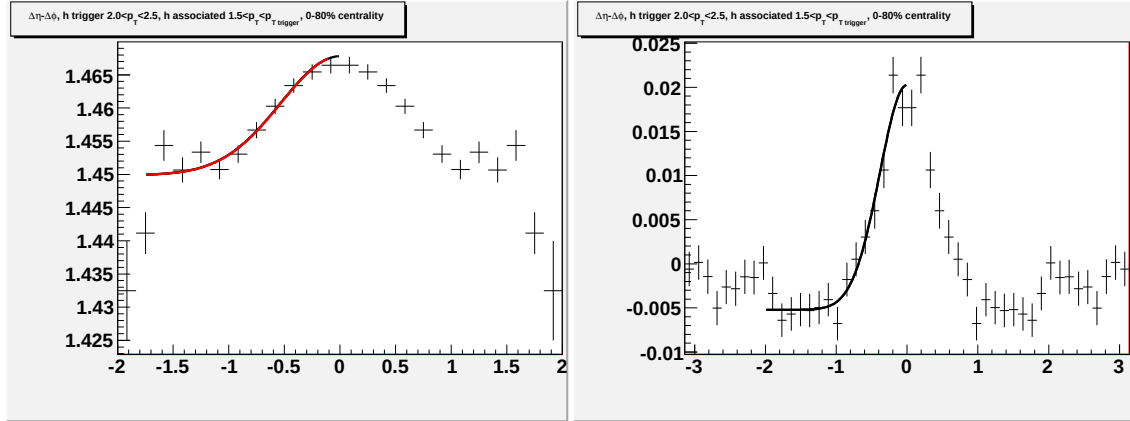


Figure F.1: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $2.0 < p_T^{trigger} < 2.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

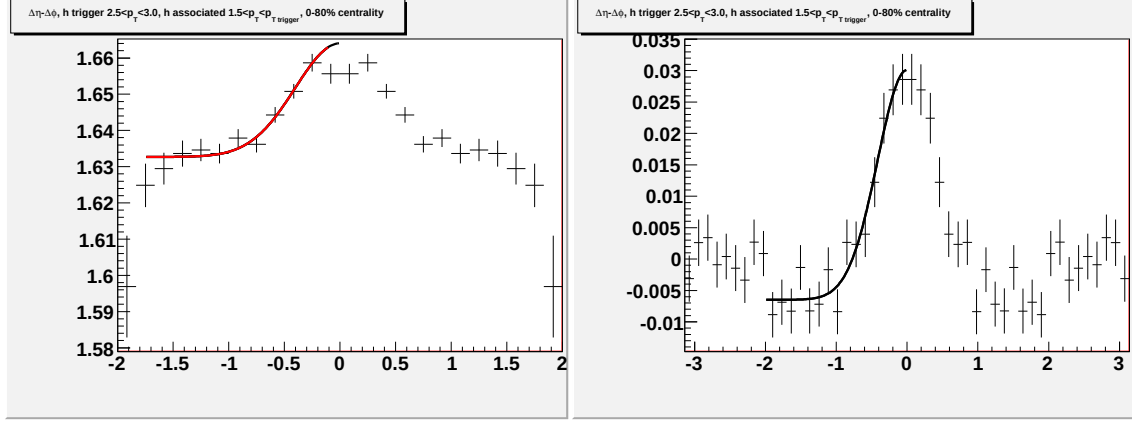


Figure F.2: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $2.5 < p_T^{trigger} < 3.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

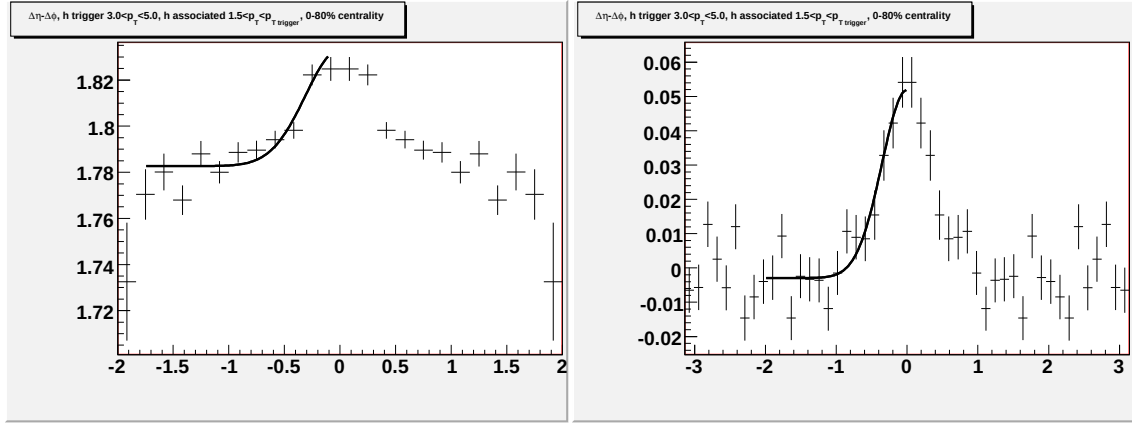


Figure F.3: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 3.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

Appendix G

Fits for $p_T^{trigger}$ dependence in $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV

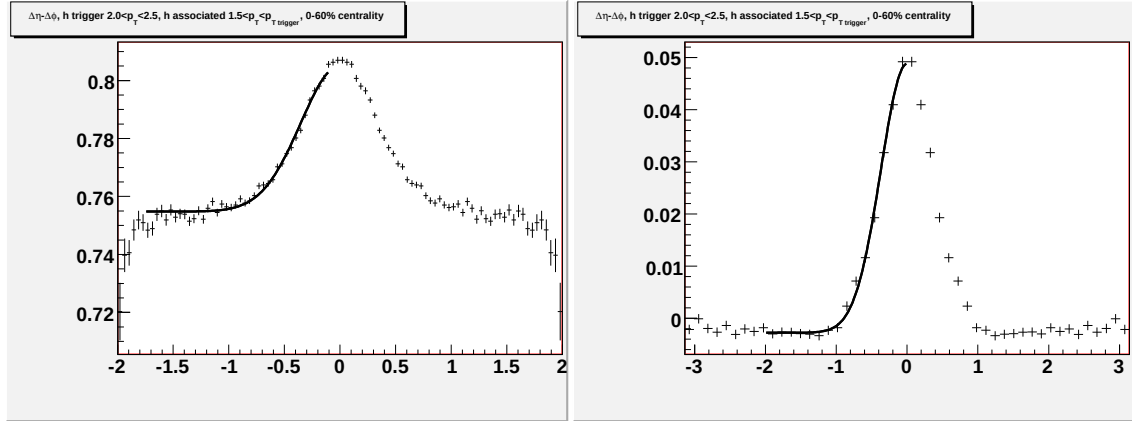


Figure G.1: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $2.0 < p_T^{trigger} < 2.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

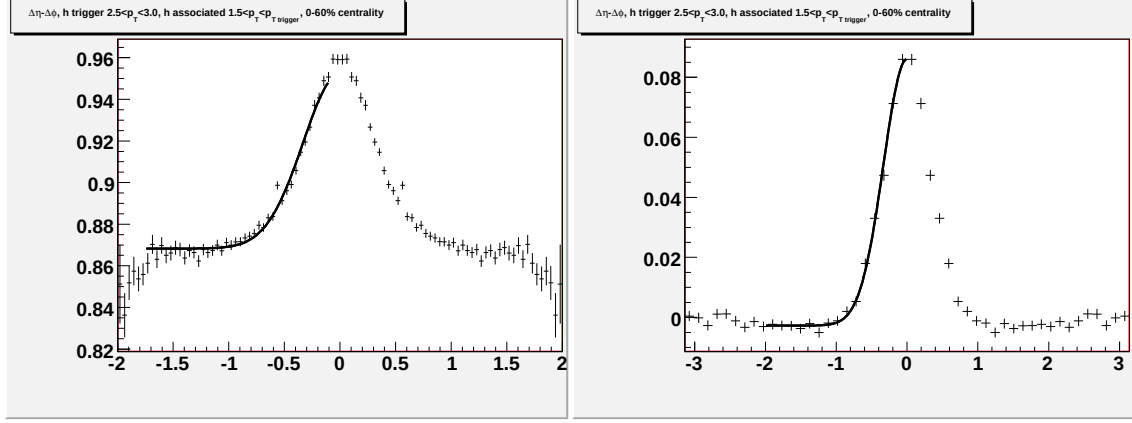


Figure G.2: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $2.5 < p_T^{trigger} < 3.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

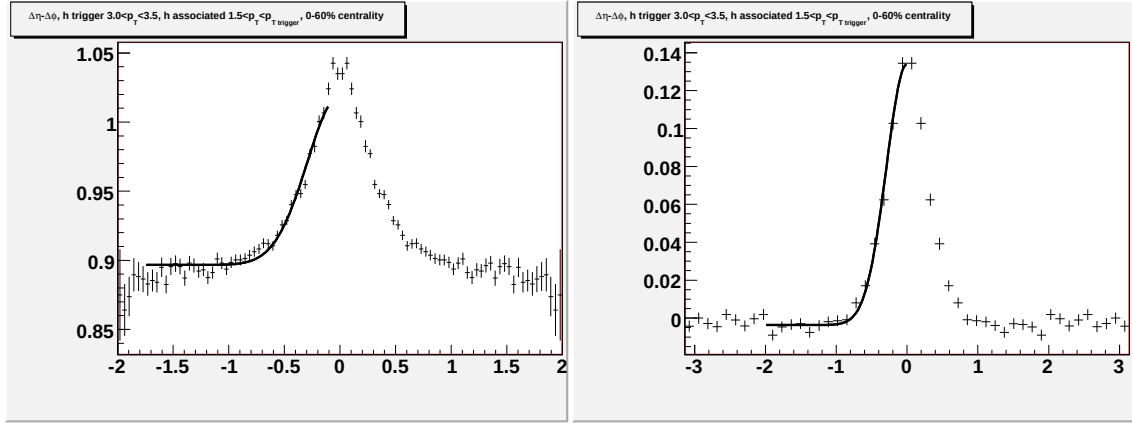


Figure G.3: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 3.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

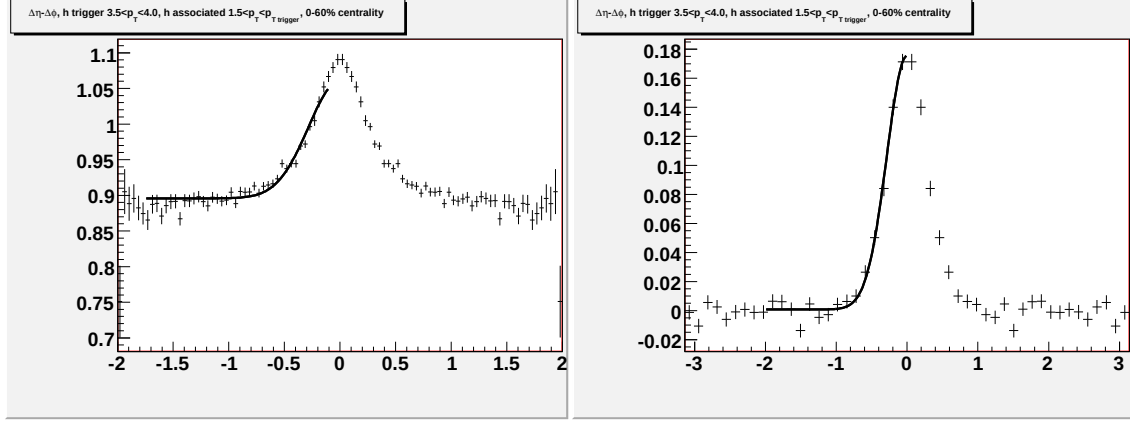


Figure G.4: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.5 < p_T^{trigger} < 4.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

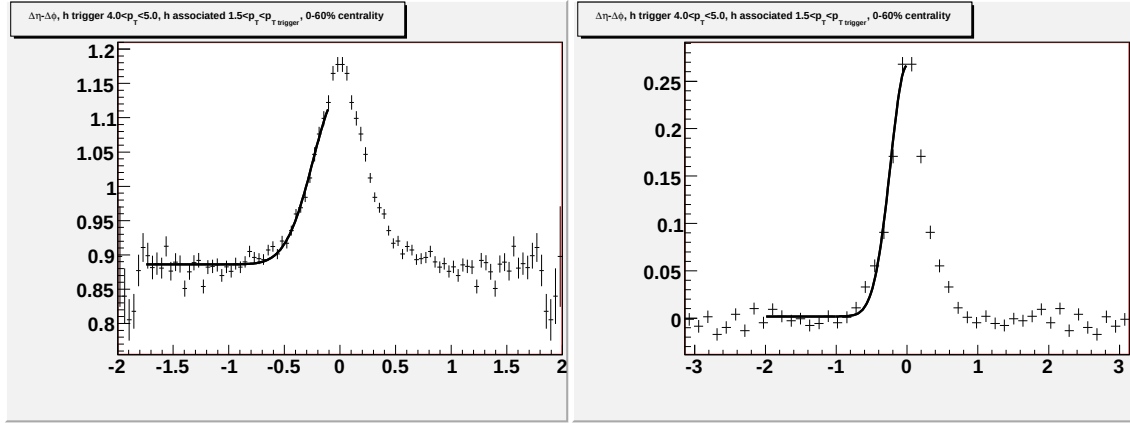


Figure G.5: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $4.0 < p_T^{trigger} < 5.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

Appendix H

Fits for $p_T^{trigger}$ dependence in
40-80% $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV

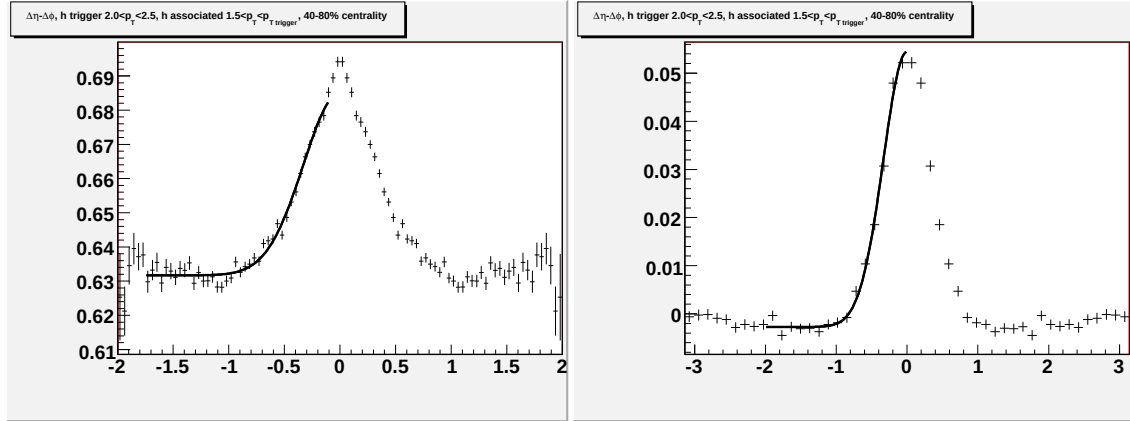


Figure H.1: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $2.0 < p_T^{trigger} < 2.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

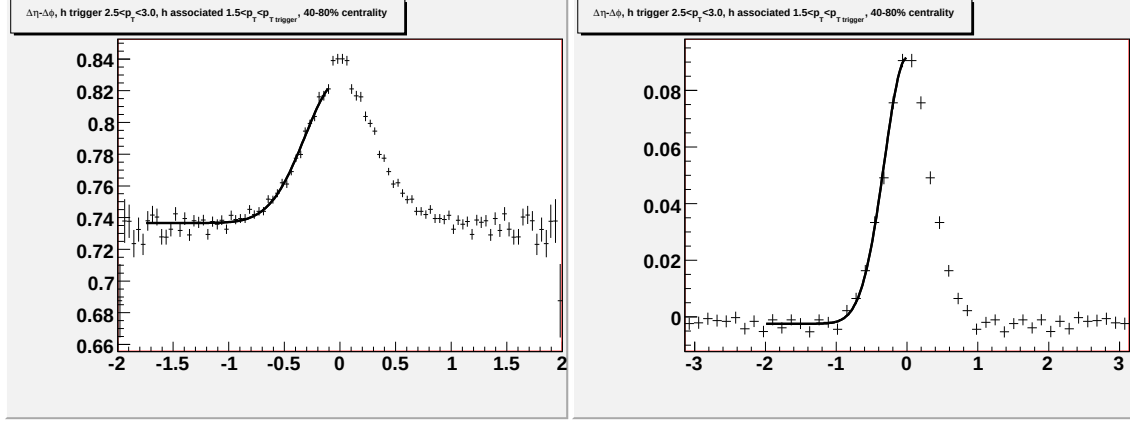


Figure H.2: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $2.5 < p_T^{trigger} < 3.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

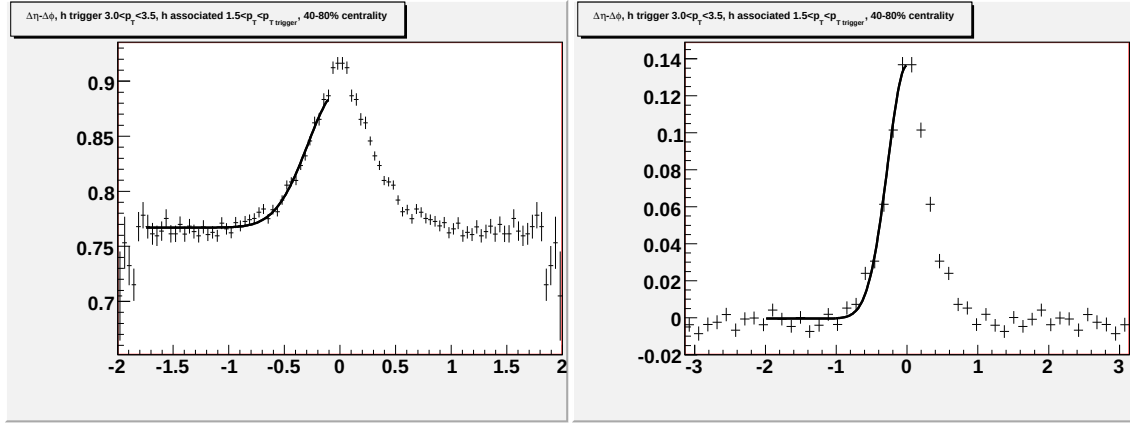


Figure H.3: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 3.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

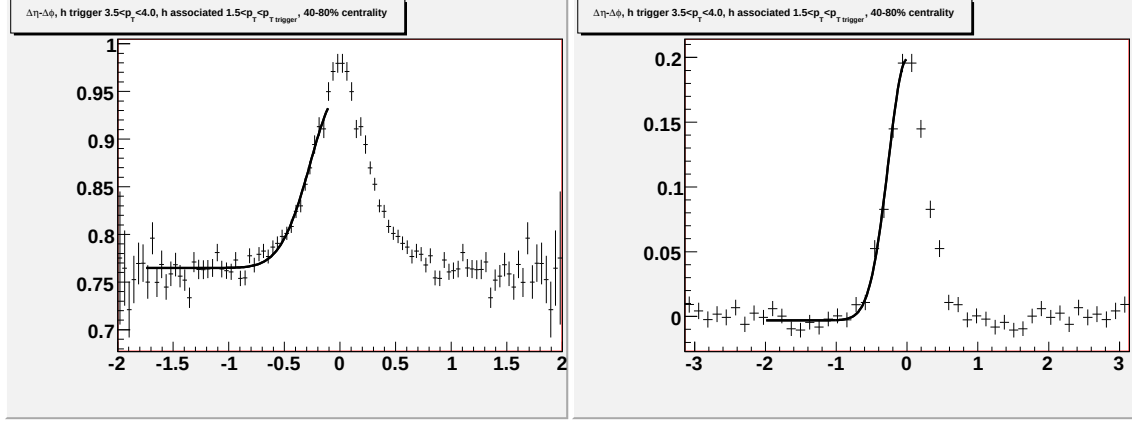


Figure H.4: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.5 < p_T^{trigger} < 4.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

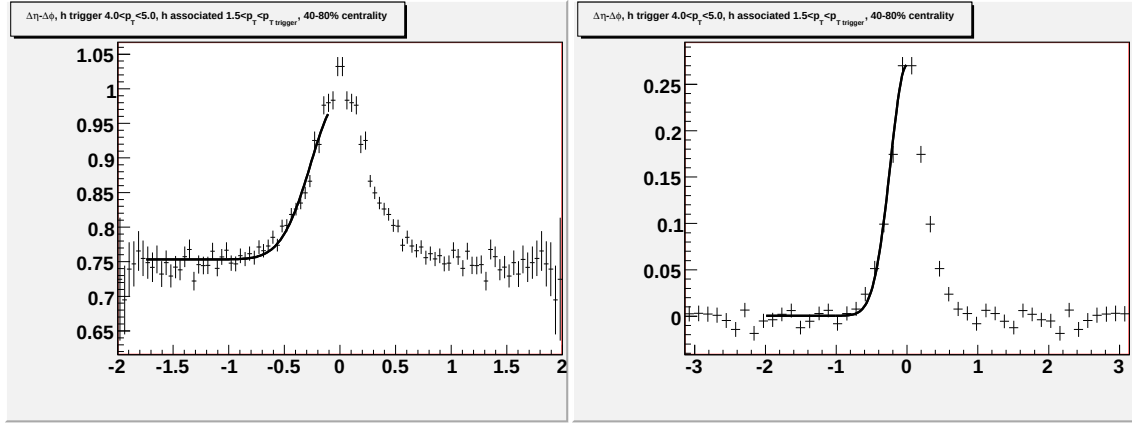


Figure H.5: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $4.0 < p_T^{trigger} < 5.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

Appendix I

Fits for $p_T^{trigger}$ dependence in $d + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV

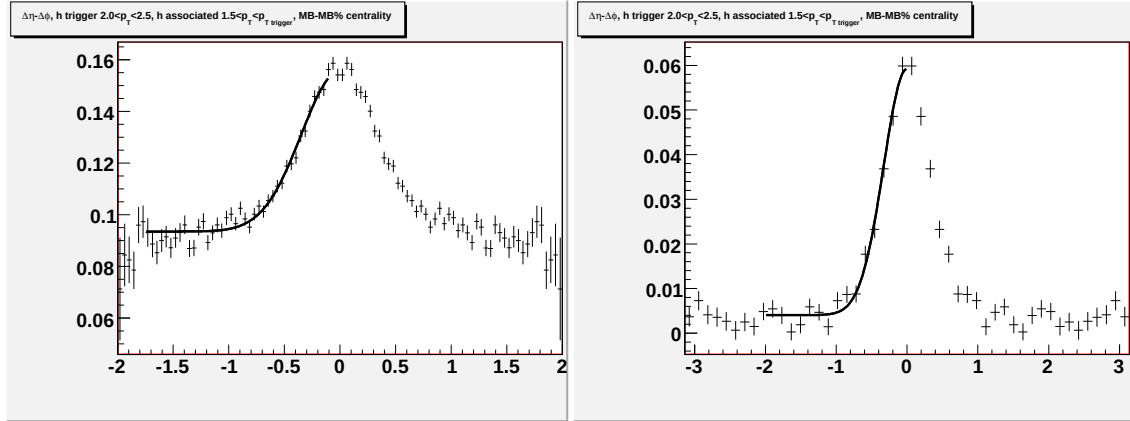


Figure I.1: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $2.0 < p_T^{trigger} < 2.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

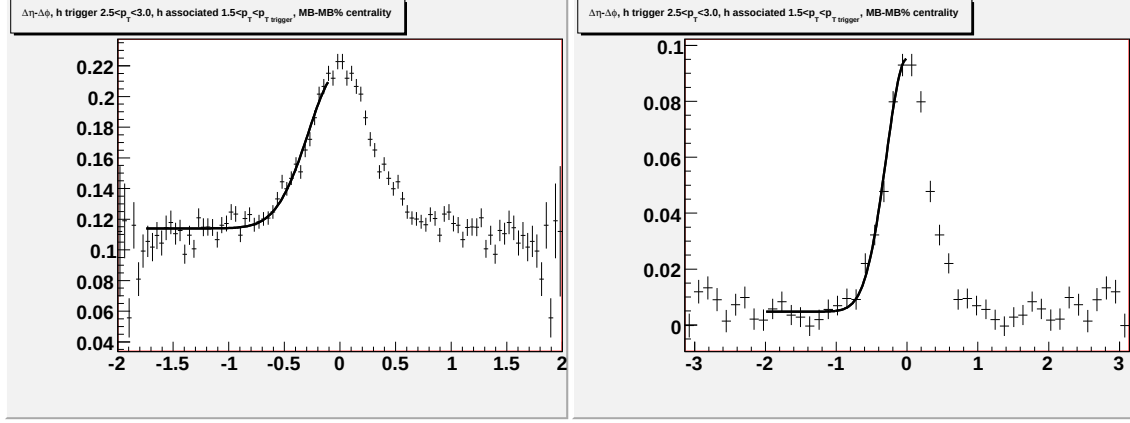


Figure I.2: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $2.5 < p_T^{trigger} < 3.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

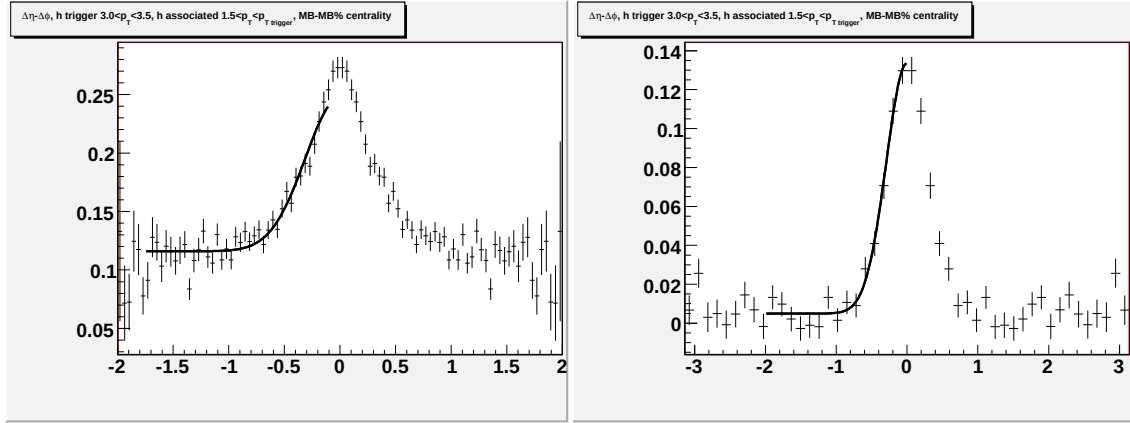


Figure I.3: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 3.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

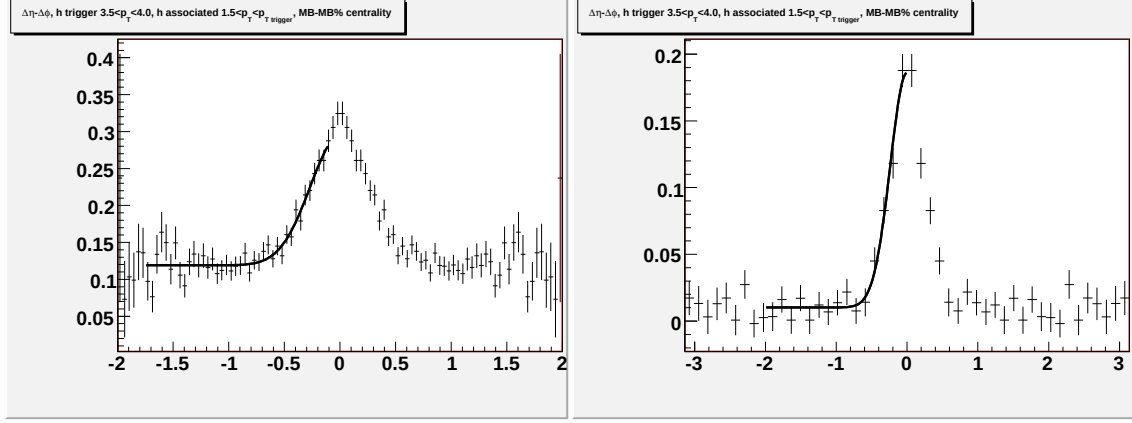


Figure I.4: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.5 < p_T^{trigger} < 4.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

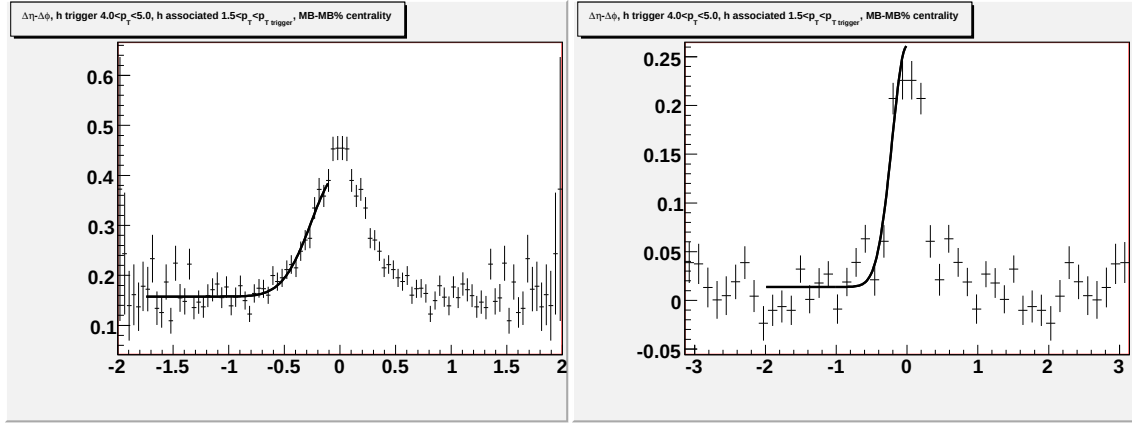


Figure I.5: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $4.0 < p_T^{trigger} < 5.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

Appendix J

N_{part} dependence for different collision energies

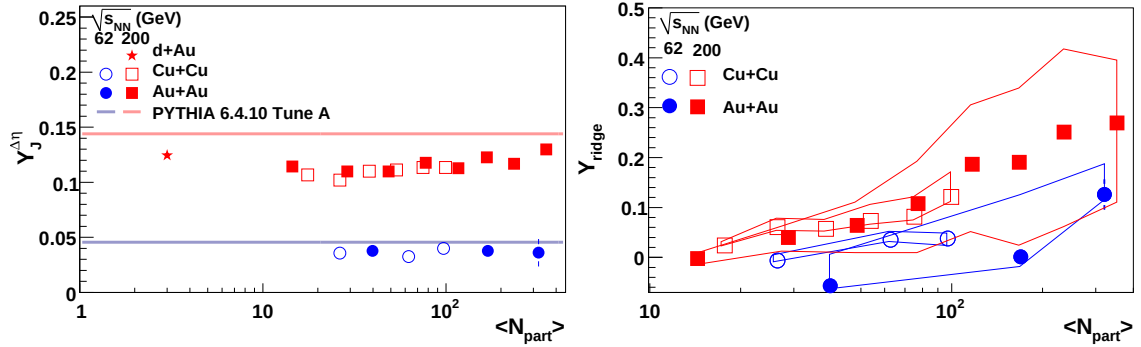


Figure J.1: N_{part} dependence of Y_{Jet} for unidentified hadrons for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 \text{ GeV/c} < p_T^{associated} < p_T^{trigger}$ for unidentified hadrons for $Cu + Cu$ and $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV and $d + Au$, $Cu + Cu$ and $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

All points are done using bin counting and the $\Delta\eta$ method except for the central $\sqrt{s_{NN}} = 62$ GeV. For this data point, the fit is used because there is clearly a residual track merging effect. Because of the residual track merging effect, the width in $\Delta\eta$ is thrown out for the width plot.

Appendix K

Fits for N_{part} dependence in $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 62$ GeV

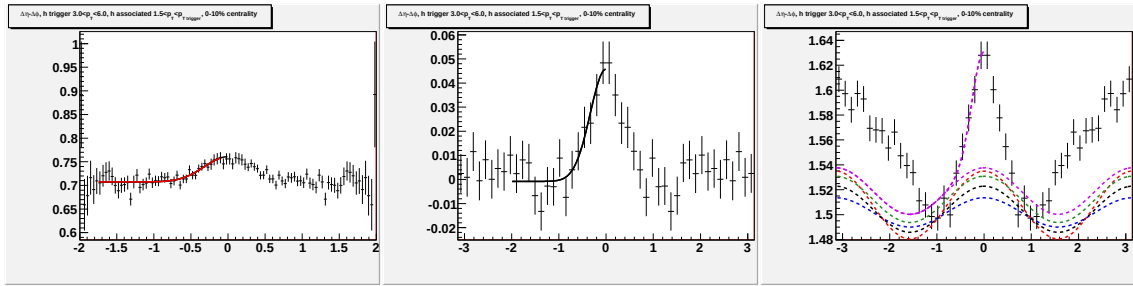


Figure K.1: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 0-10% $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 62$ GeV.

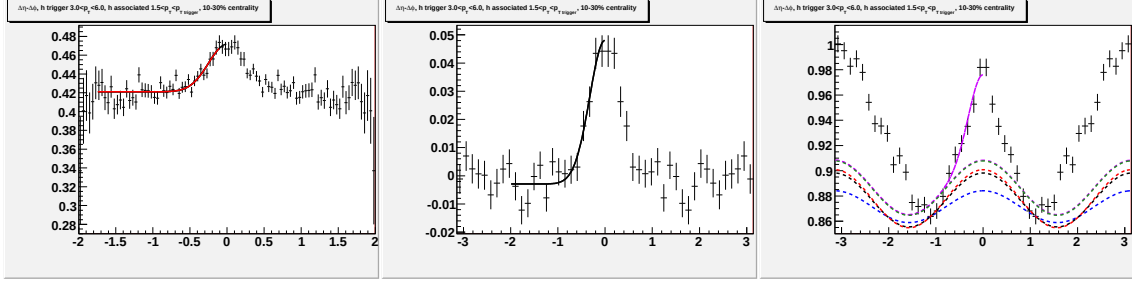


Figure K.2: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 10-30% $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 62$ GeV.

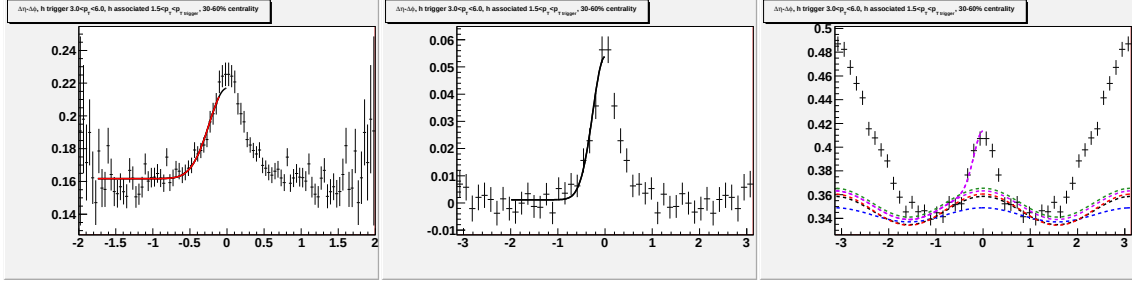


Figure K.3: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 30-60% $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 62$ GeV.

Appendix L

Fits for N_{part} dependence in $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV

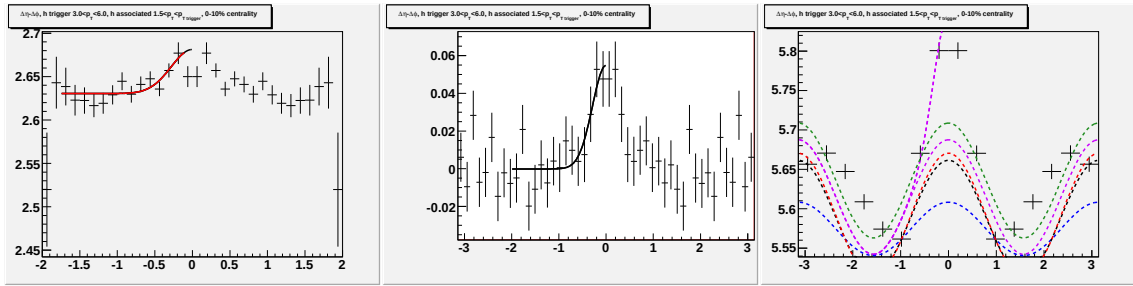


Figure L.1: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 0-10% $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV.

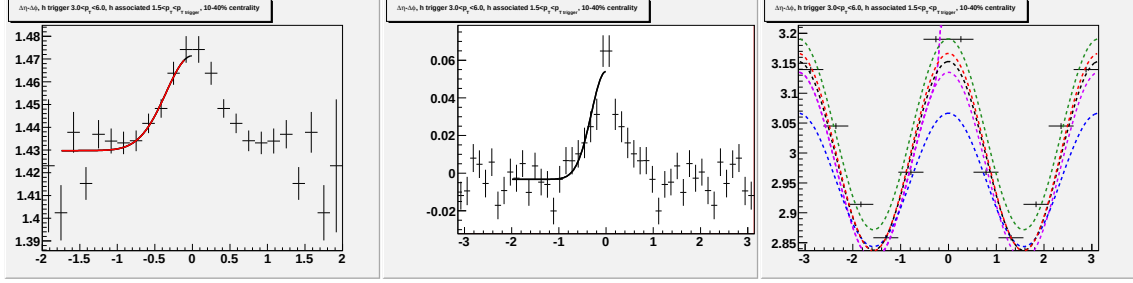


Figure L.2: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 10-40% $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV.

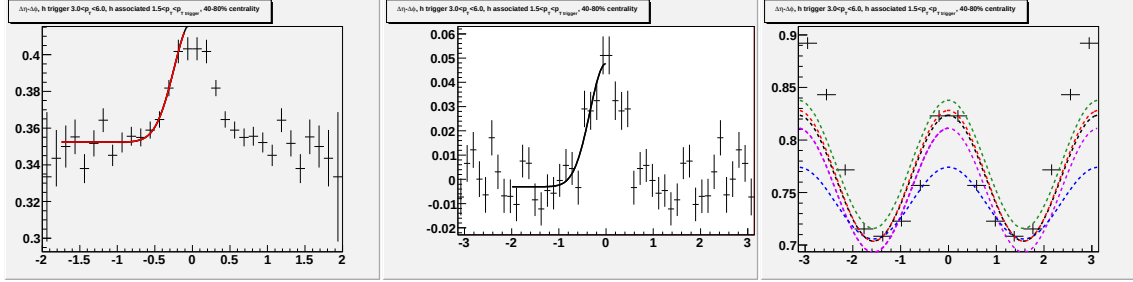


Figure L.3: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 40-80% $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV.

Appendix M

Fits for N_{part} dependence in $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV

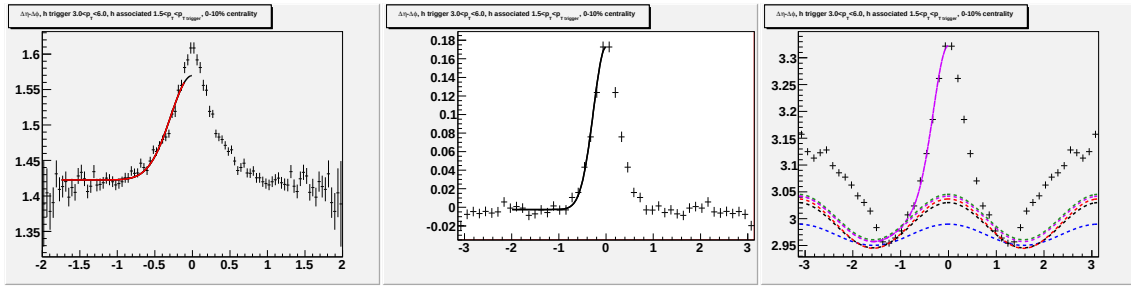


Figure M.1: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 0-10% $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

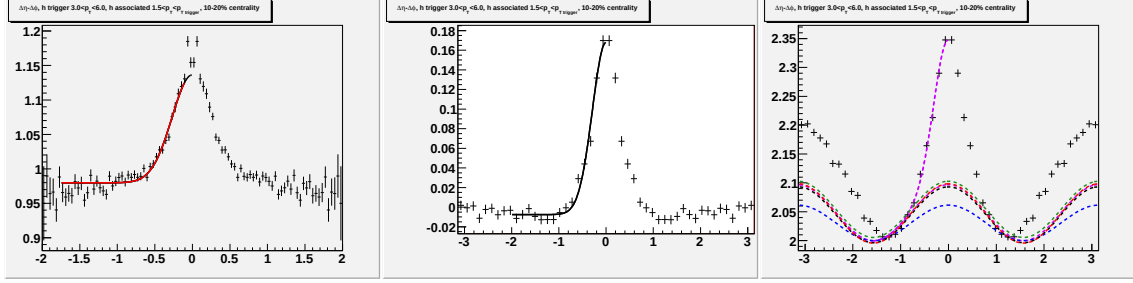


Figure M.2: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 10-20% $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

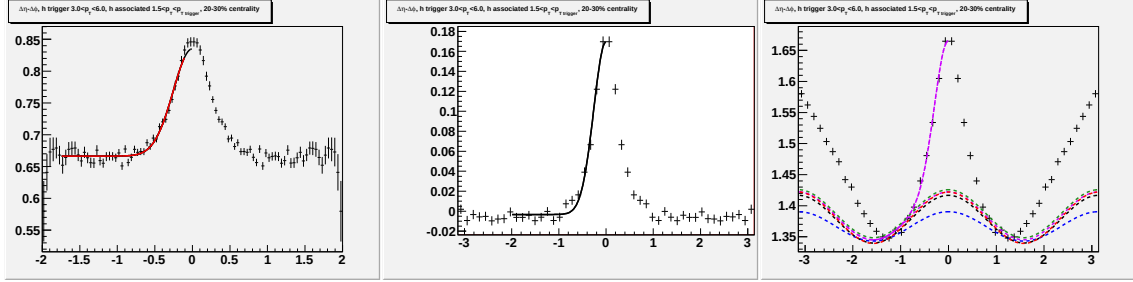


Figure M.3: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 20-30% $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

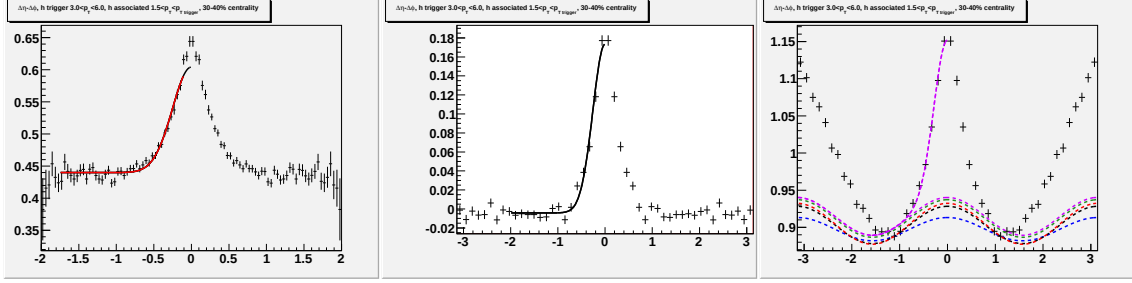


Figure M.4: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 30-40% $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

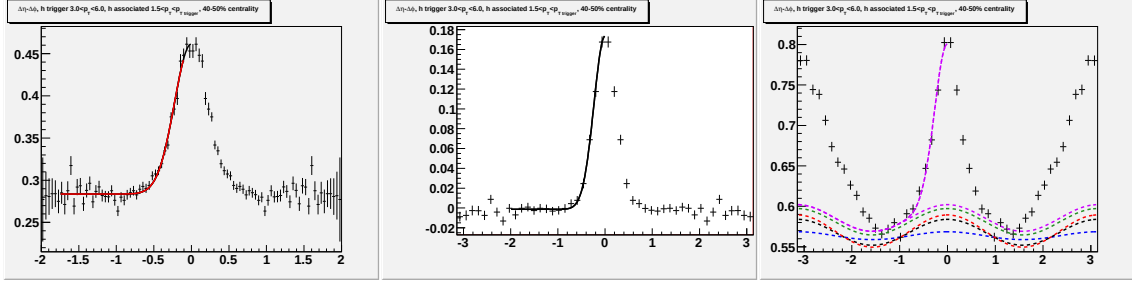


Figure M.5: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 40-50% $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

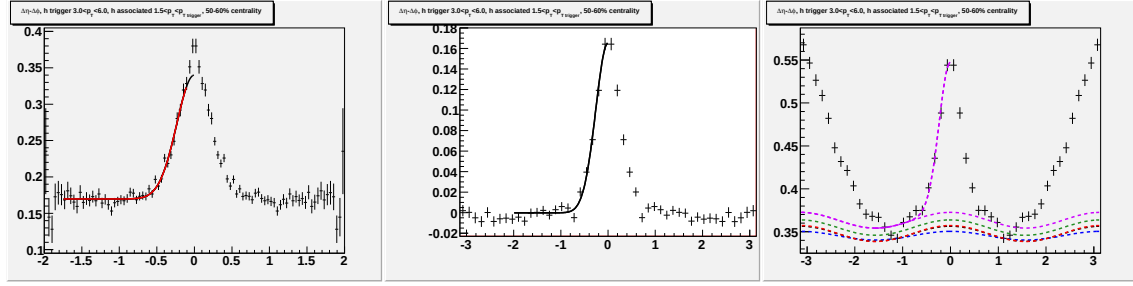


Figure M.6: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 50-60% $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

Appendix N

Fits for N_{part} dependence in $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV

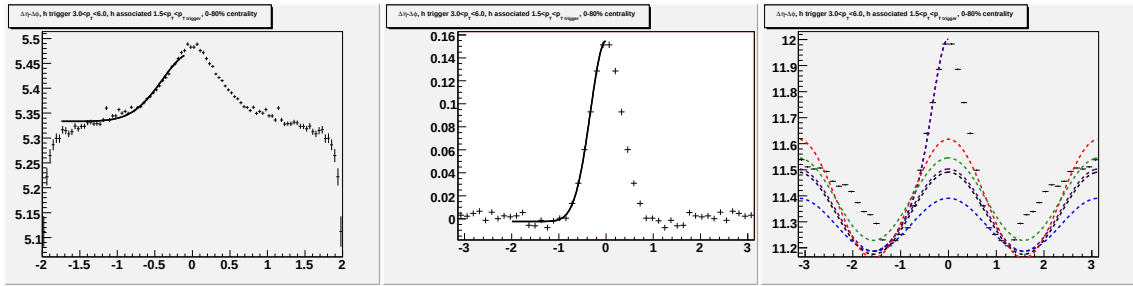


Figure N.1: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 0-12% $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

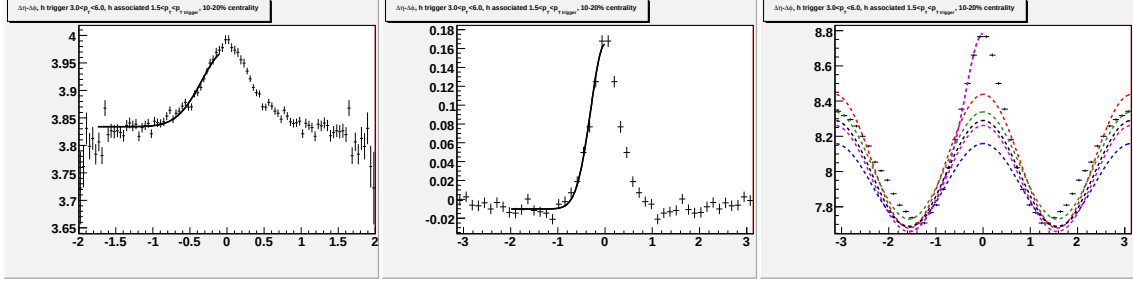


Figure N.2: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 10-20% $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

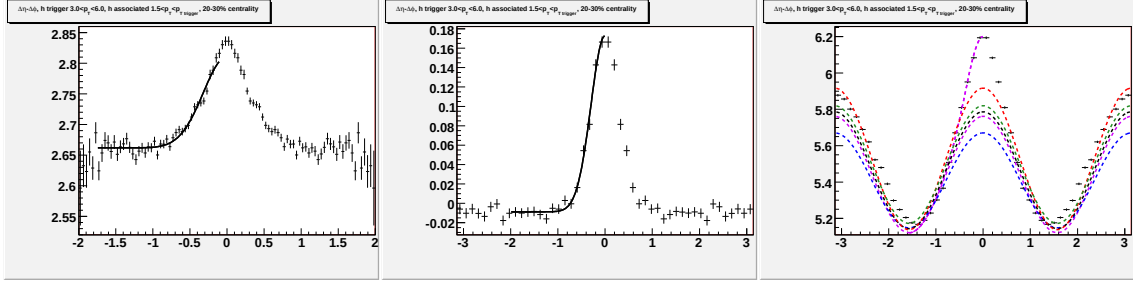


Figure N.3: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 20-30% $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

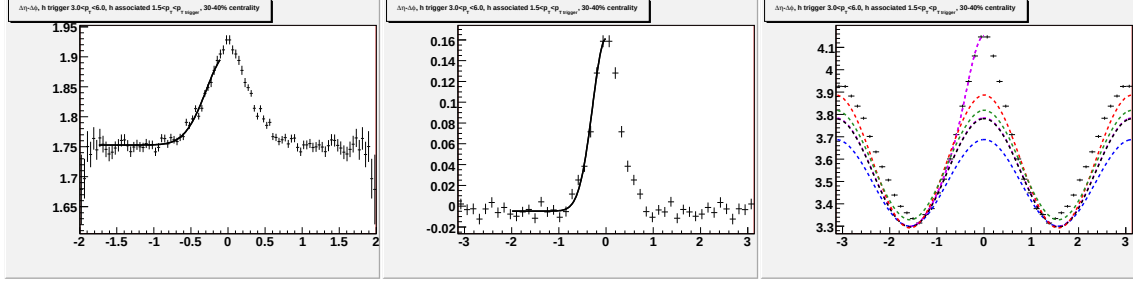


Figure N.4: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 30-40% $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

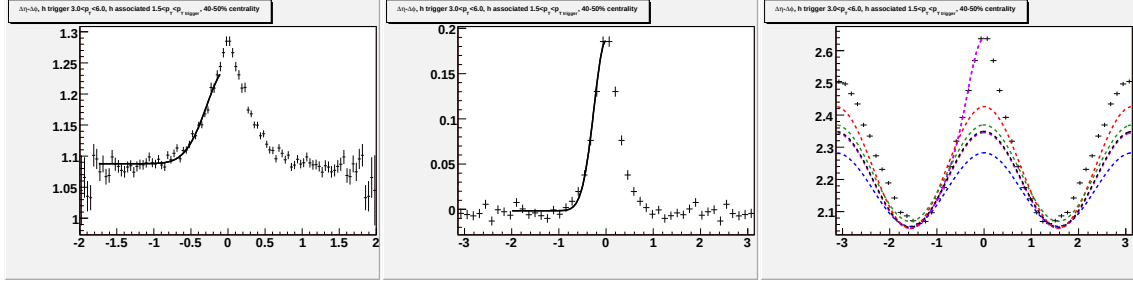


Figure N.5: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 40-50% $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

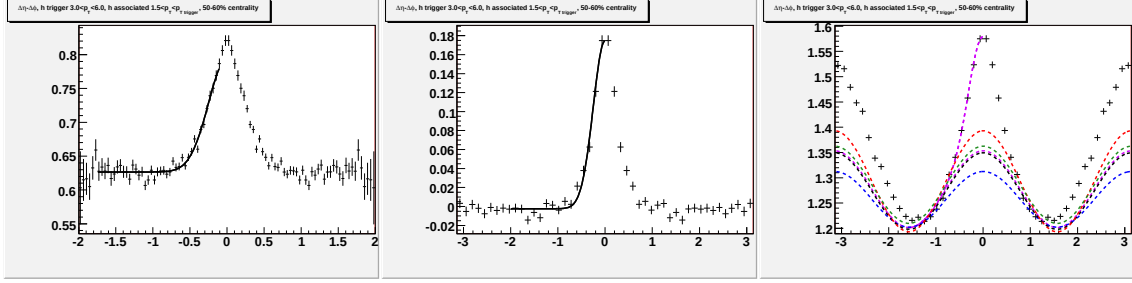


Figure N.6: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 50-60% $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

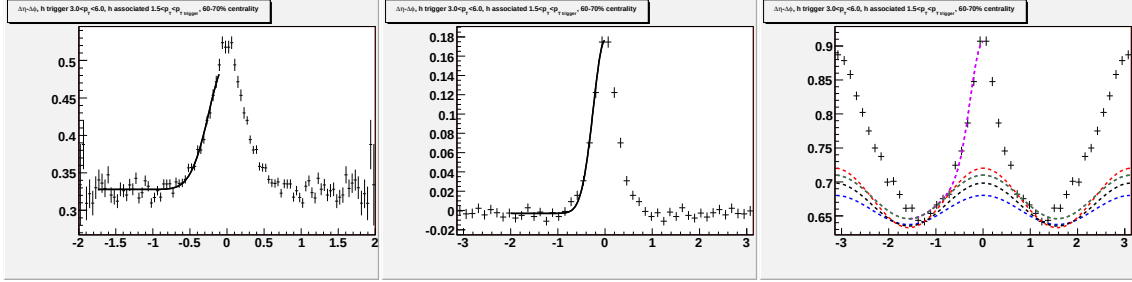


Figure N.7: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 60-70% $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

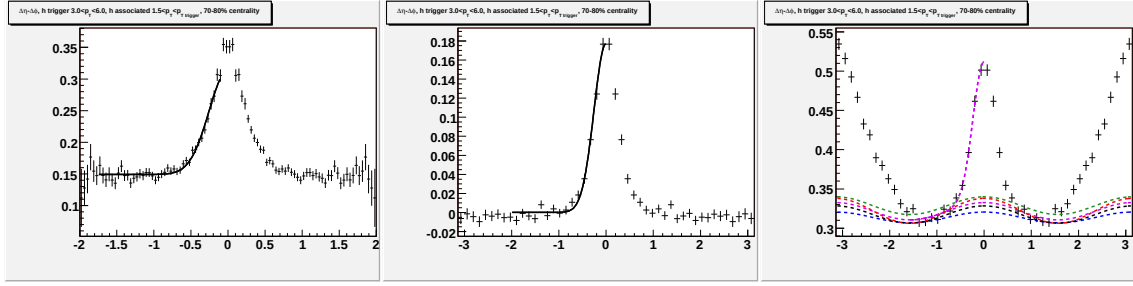


Figure N.8: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 70-80% Au + Au collisions at $\sqrt{s_{NN}} = 200$ GeV.

Appendix O

Fits for N_{part} dependence in $d + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV

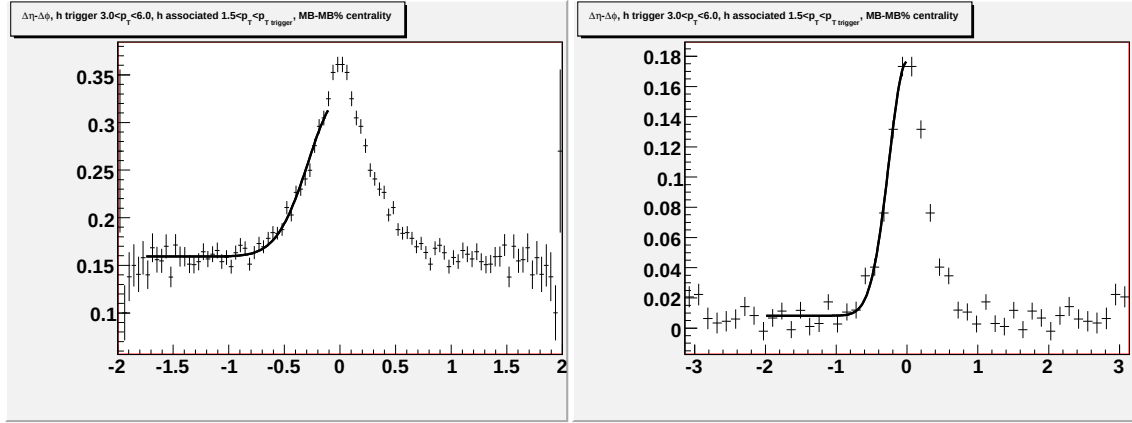


Figure O.1: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for minimum bias $d + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

Appendix P

$p_T^{associated}$ dependence for different collision energies

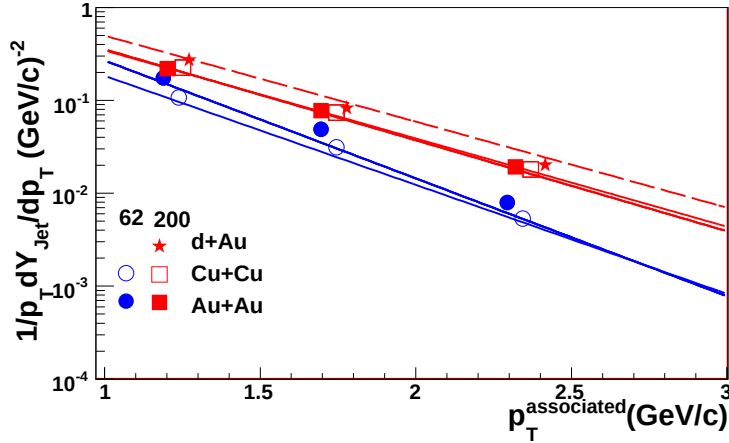


Figure P.1: $p_T^{associated}$ dependence of Y_{Jet} for $1.5 \text{ GeV/c} < p_T^{associated} < p_T^{trigger}$ for $3.0 < p_T^{trigger} < 6.0 \text{ GeV/c}$ for 0-60% central $Cu + Cu$ and 0-80% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 62 \text{ GeV}$ and minimum bias $d + Au$, 0-60% central $Cu + Cu$ and 0-12% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200 \text{ GeV}$.

This figure still needs updated dAu data for both the yields and the widths. All points are done using bin counting and the $\Delta\eta$ method.

Appendix Q

Fits for $p_T^{associated}$ dependence in $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 62$ GeV

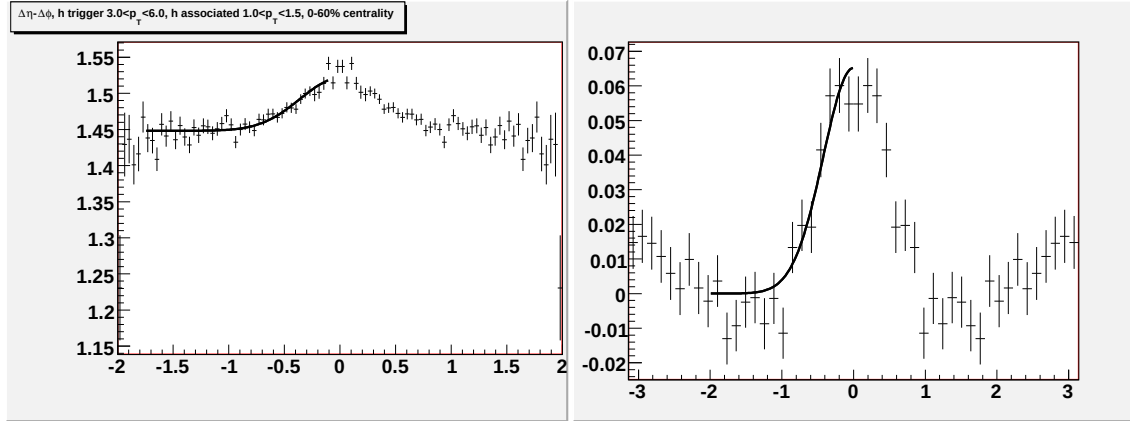


Figure Q.1: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.0 < p_T^{associated} < 1.5$ GeV/c.

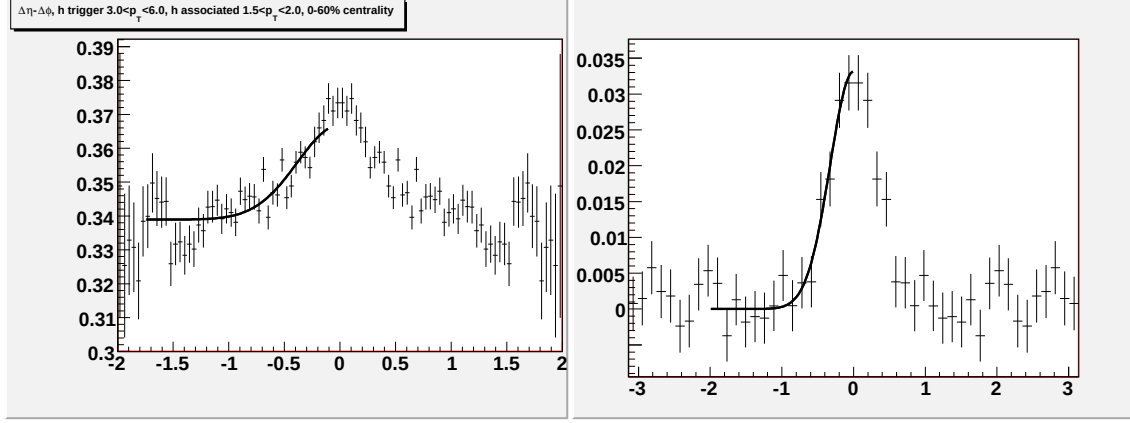


Figure Q.2: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < 2.0$ GeV/c.

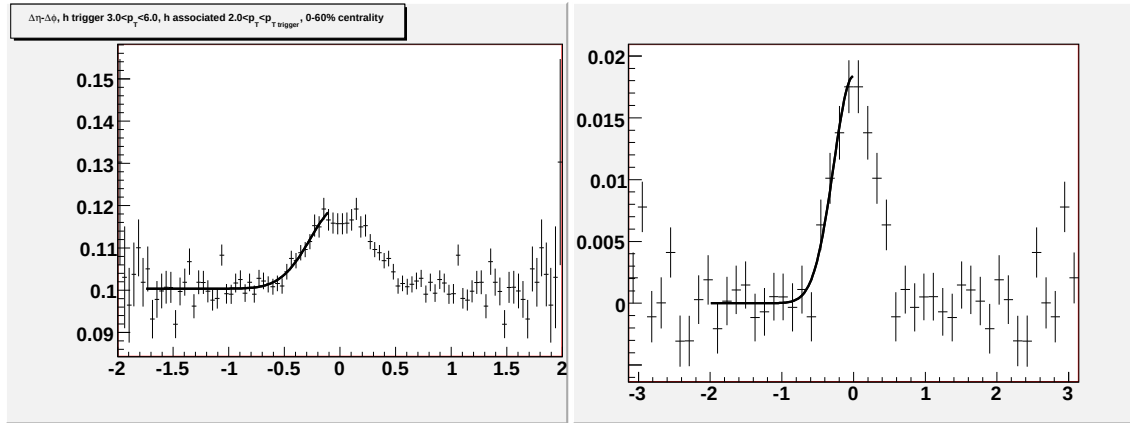


Figure Q.3: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $2.0 < p_T^{associated} < 3.0$ GeV/c.

Appendix R

Fits for $p_T^{associated}$ dependence in $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV

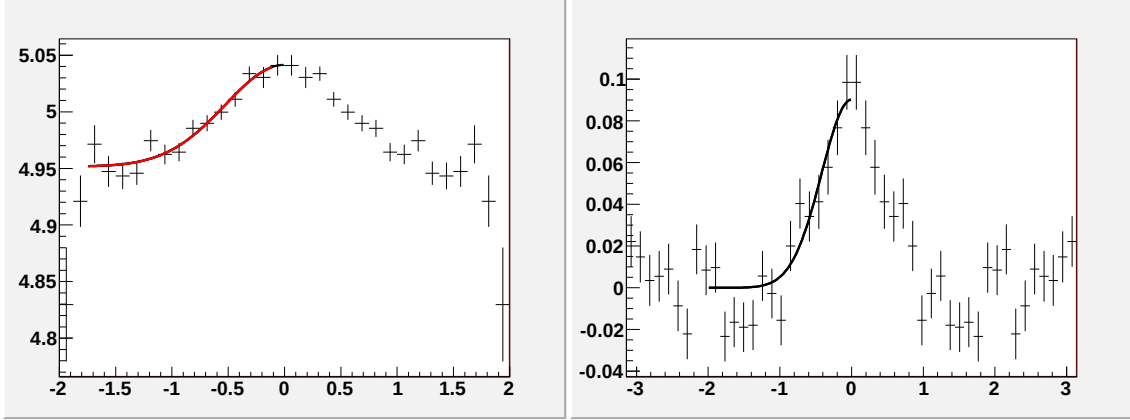


Figure R.1: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.0 < p_T^{associated} < 1.5$ GeV/c.

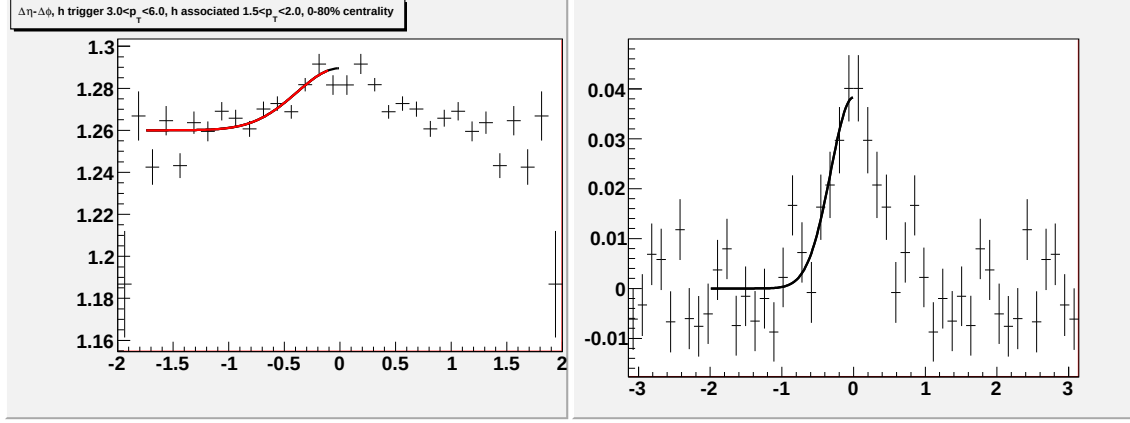


Figure R.2: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < 2.0$ GeV/c.

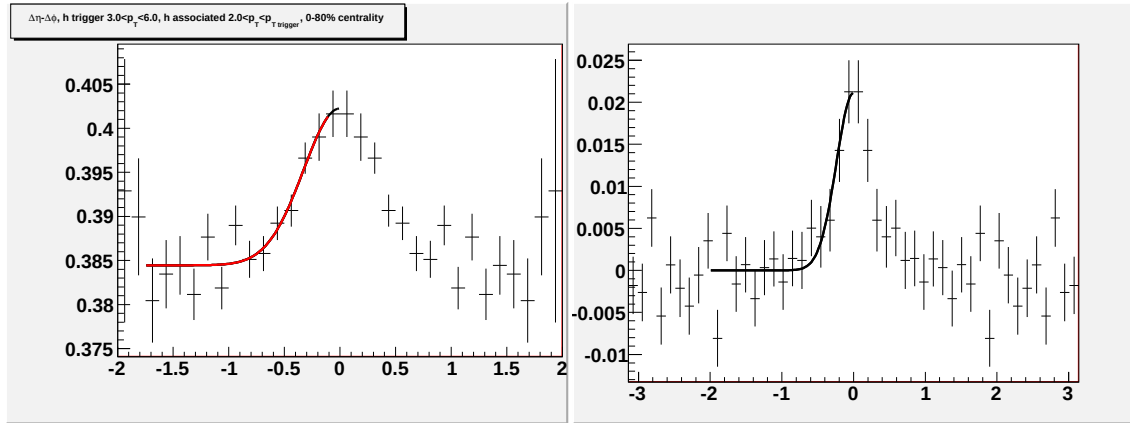


Figure R.3: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $2.0 < p_T^{associated} < 3.0$ GeV/c.

Appendix S

Fits for $p_T^{associated}$ dependence in $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV

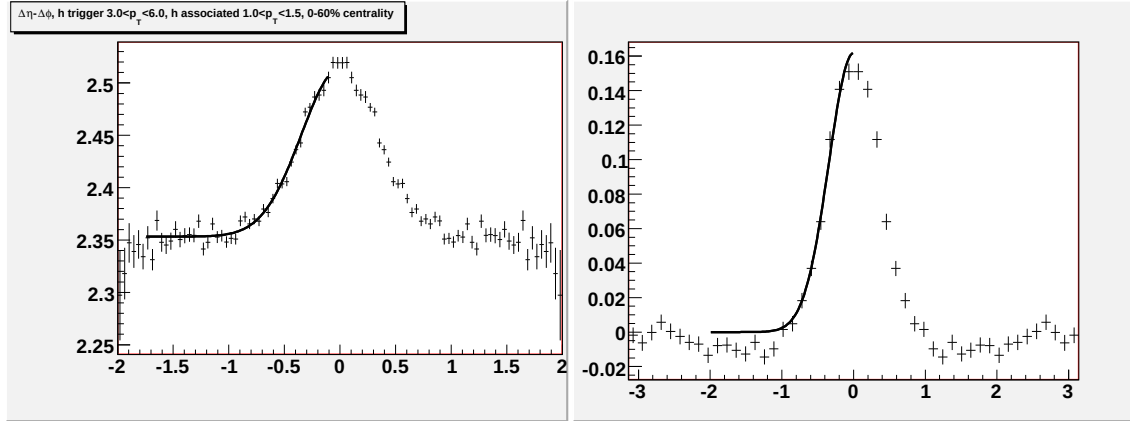


Figure S.1: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.0 < p_T^{associated} < 1.5$ GeV/c.

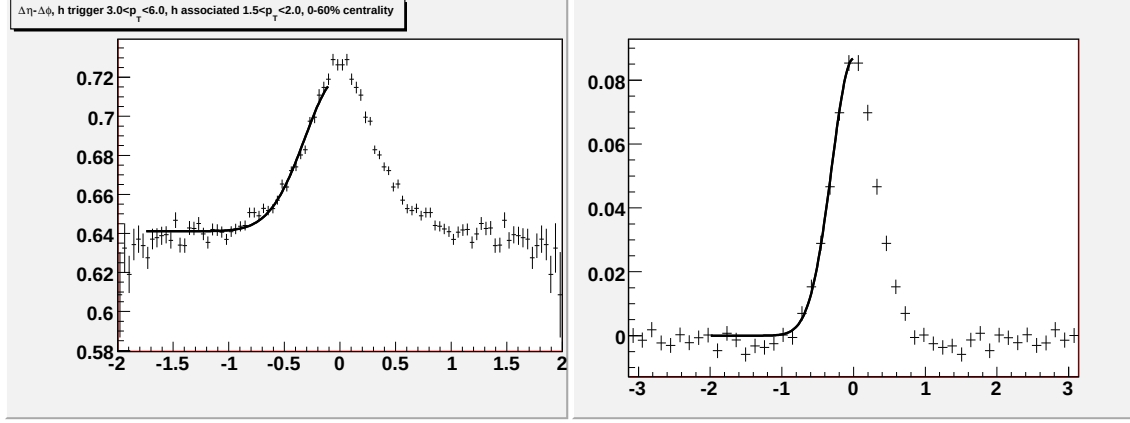


Figure S.2: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < 2.0$ GeV/c.

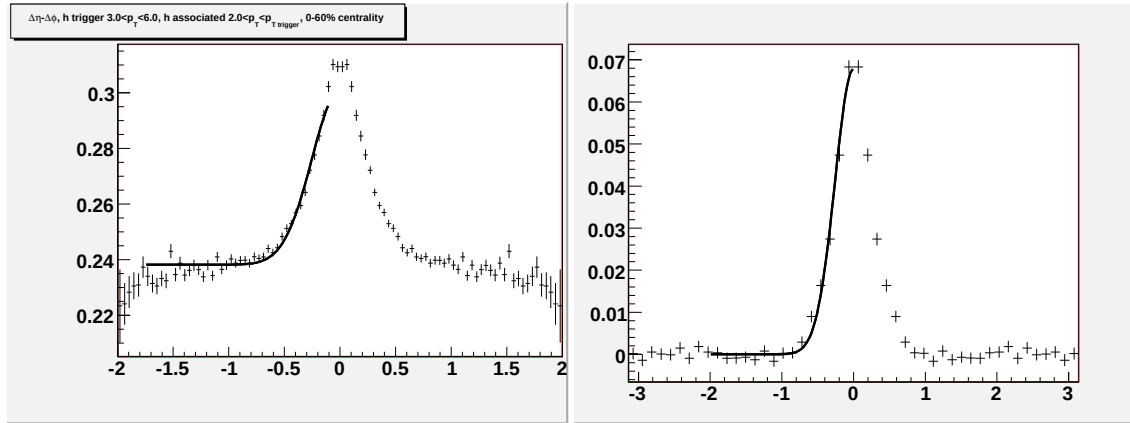


Figure S.3: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $2.0 < p_T^{associated} < 3.0$ GeV/c.

Appendix T

Fits for $p_T^{associated}$ dependence in
minimum bias $Au + Au$ collisions at
 $\sqrt{s_{NN}} = 200 \text{ GeV}$

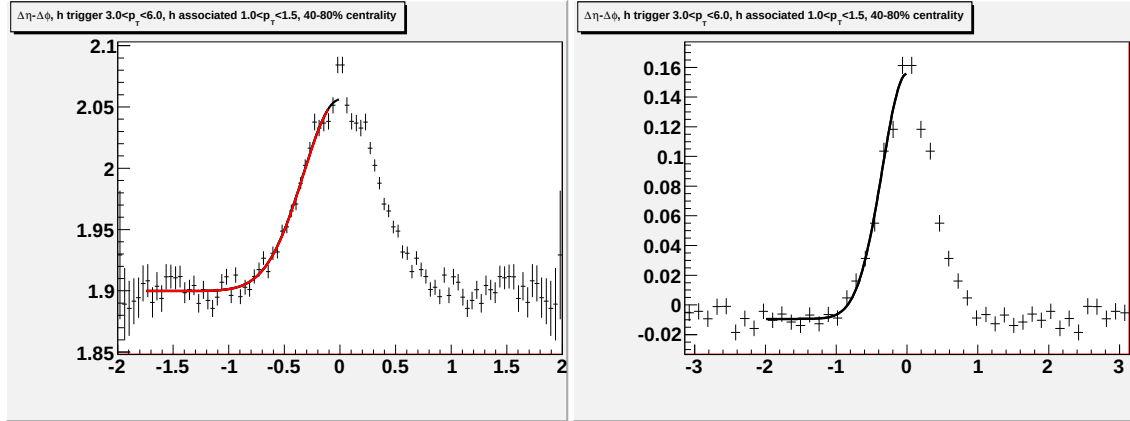


Figure T.1: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0 \text{ GeV/c}$ and $1.0 < p_T^{associated} < 1.5 \text{ GeV/c}$.

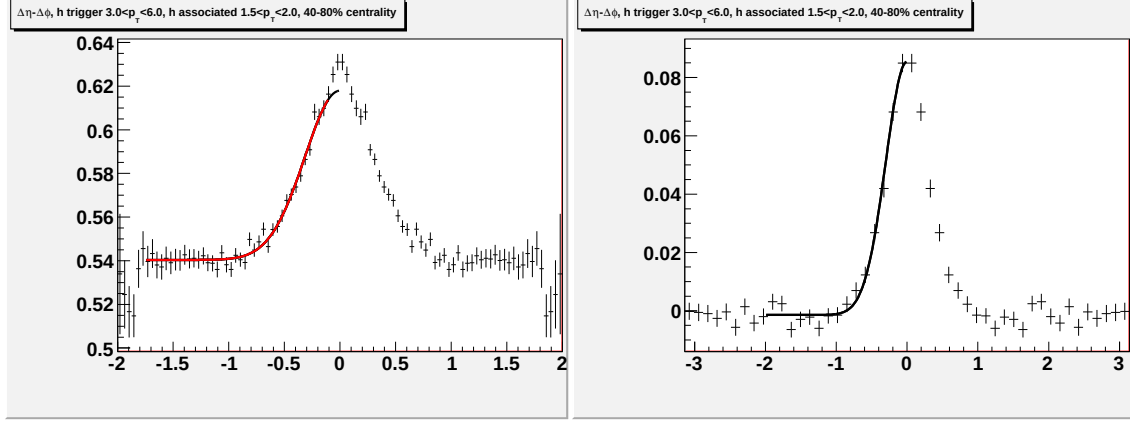


Figure T.2: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < 2.0$ GeV/c.

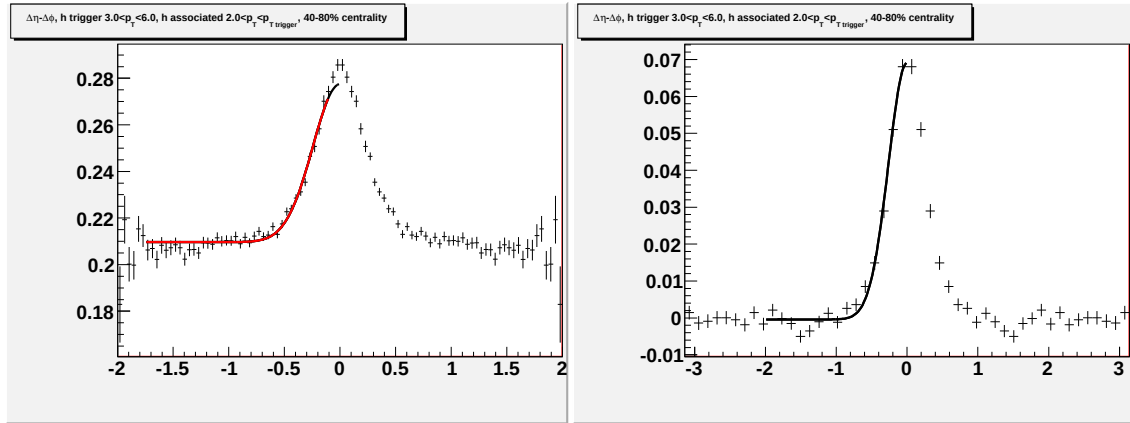


Figure T.3: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $2.0 < p_T^{associated} < 3.0$ GeV/c.

Appendix U

Fits for $p_T^{associated}$ dependence in $d + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV

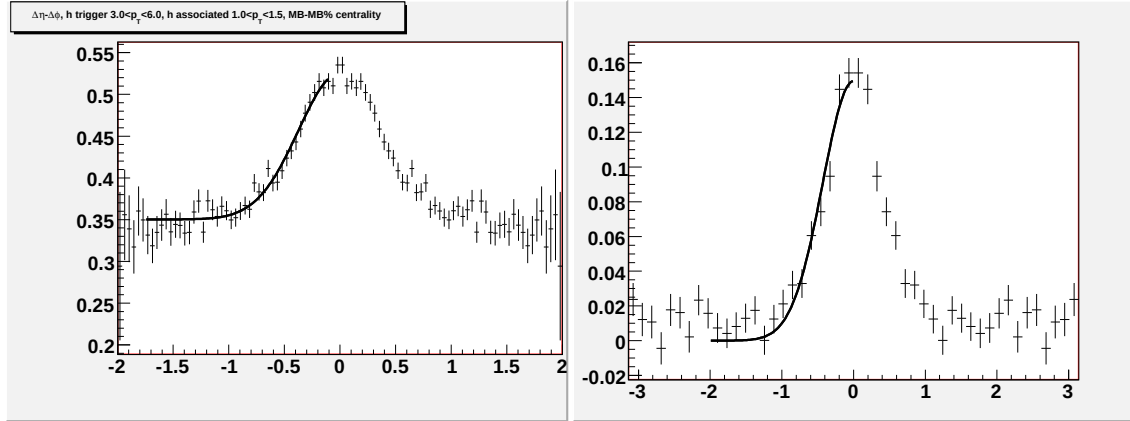


Figure U.1: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.0 < p_T^{associated} < 1.5$ GeV/c.

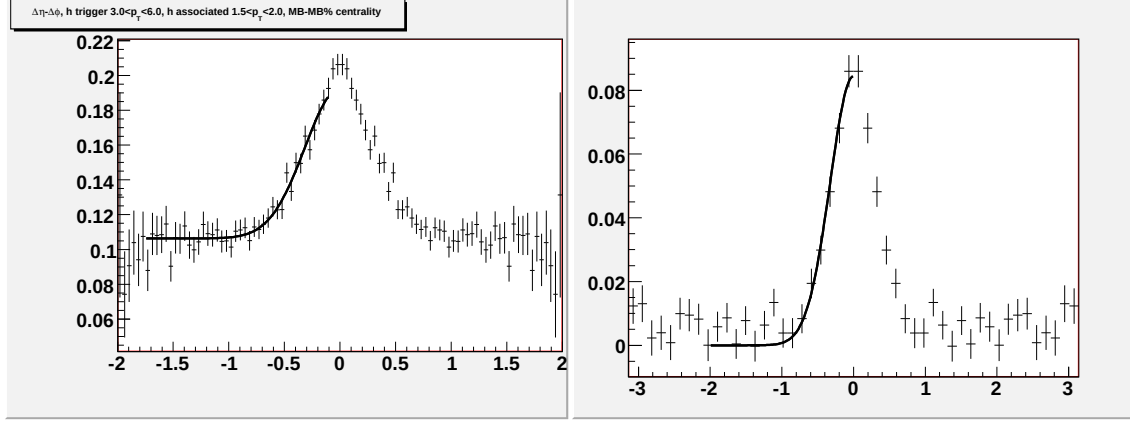


Figure U.2: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < 2.0$ GeV/c.

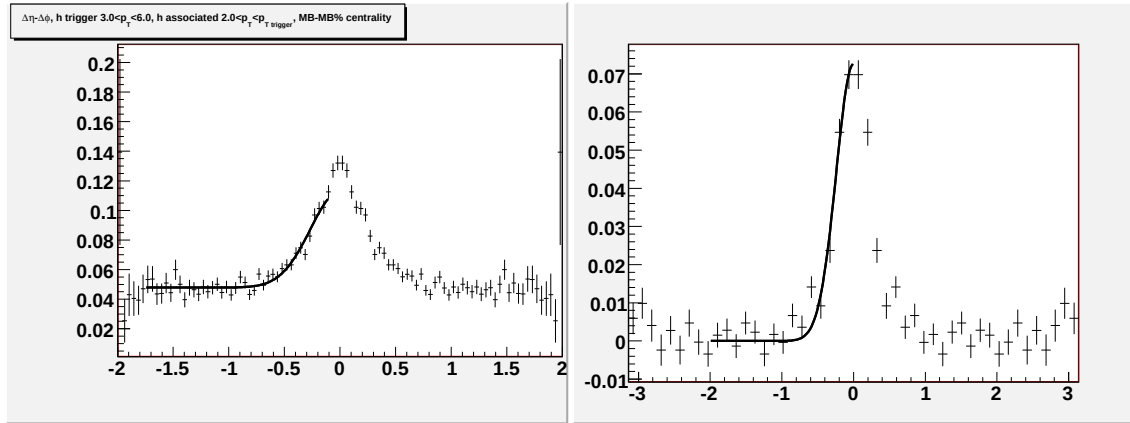


Figure U.3: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $2.0 < p_T^{associated} < 3.0$ GeV/c.

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